ashika
Institutional Equity Research

Azad Engineering

Initiating Coverage

Research Analyst: Koushik Mohan & Juili Baviskar

Capital Goods Sector | Initiating Coverage

Azad Engineering (AZAD)

BUY: 49%| Target Price: Rs. 2,193

ashika Institutional Equity Research

25 March 2026

Precision-Driven Moat with Multi-Year Growth Visibility, Initiating Coverage with BUY

Key Triggers:

- Revenue grew at ~30% CAGR (FY22–FY25) and is expected to sustain 30%+ growth, supported by a strong order book and rising Aerospace & Defence mix (~50% medium term); EBITDA margin expanded from ~32% to ~36%, driven by indigenisation and operating leverage.
- Azad operates in large and growing markets - Aerospace & Defence TAM (~Rs90 bn in 2022 to ~Rs153bn by 2027; ~9% CAGR) and Energy Turbines (~Rs1,280bn to Rs1,810bn; ~7% CAGR). Supported by steady Oil & Gas demand, it is well placed to scale wallet share from ~1–2% to ~3–4%.
- Dedicated lean facilities under Phase-1 and planned Phase-2 capacity expansion enhance OEM integration, improve efficiency and support execution of a growing order pipeline, with blended asset turnover at ~1.8x.
- The company is gradually **moving up the turbine value chain** by expanding from compressor airfoils into combustion components and propulsion systems, increasing wallet share per platform and strengthening its role within global turbine ecosystems.
- Entry into end-to-end propulsion (GTRE) and participation in nuclear/advanced turbine programs add long-term growth optionality and a diversified revenue stream

Valuation & Outlook:

Backed by strong revenue growth visibility, Azad is well positioned to sustain its growth momentum, supported by a robust order book and long-term contracts with global OEMs. The company operates in a niche segment of manufacturing precision, mission-critical and zero-defect components, where lengthy qualification cycles create high entry barriers and ensure strong customer stickiness besides providing multi-year revenue visibility. Additionally, commissioning of dedicated lean facilities, ongoing capex plans aligned with confirmed demand and structural cost advantages of ~30% over global peers further strengthen its competitive positioning and scalability.

At CMP of Rs1,471, Azad trades at ~40x FY28E EPS. We initiate coverage with a BUY and Target Price of Rs2,193, valuing it at ~60x FY28E EPS. We believe this valuation is justified given Azad's strong execution track record, high-margin precision manufacturing capabilities, deep integration with global OEM supply chains and long runway for growth, driven by Aerospace, Defence and Energy sectors.

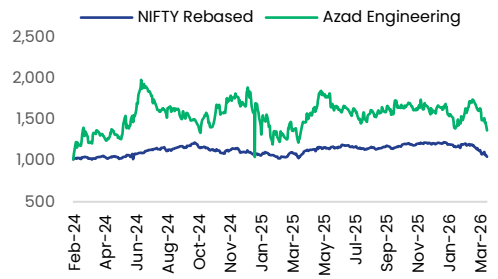
Stock Information

CMP (Rs)	1,471
Market Cap (Rs bn)	95
Free Float (%)	44
52 Week H/L	1,899/1,128
O/S Shares (mn)	64.6
3M Daily Avg Volume (mn)	0.19

Shareholding Pattern (%)

	Jun-25	Sep-25	Dec-25
Promoters	55.4	55.4	55.8
FII / NRI	16.2	15.8	15.3
MF/DII	8.9	9.8	10.7
Others	19.5	19.0	18.1

Stock Relative Performance



Stock Performance	3M	6M	12M
Absolute Return	(17.5)	(15.8)	(7.7)

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Financials (Rs mn)	FY24	FY25	FY26E	FY27E	FY28E
Revenue	3,408	4,574	6,174	8,027	10,836
EBITDA	1,166	1,613	2,223	3,010	4,118
PAT	578	865	1,352	1,644	2,362
CFO	(69)	537	510	867	1,025
EPS (Rs)	9.8	13.4	20.9	25.5	36.6

Key Ratios	FY24	FY25	FY26E	FY27E	FY28E
P/E	151.6	110.6	70.8	58.2	40.5
EV/EBITDA	75.0	56.5	43.0	31.8	23.5
RoE (%)	9.0	6.2	8.8	9.7	12.2
RoCE (%)	10.7	5.7	7.4	9.3	11.7
WC days	271	265	291	292	290

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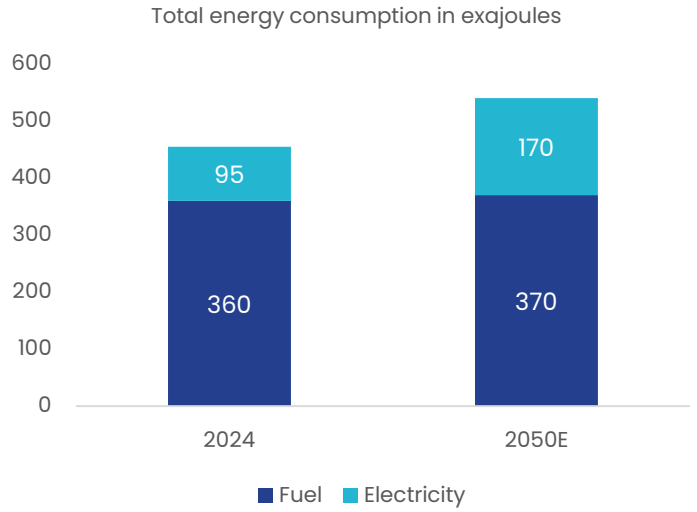
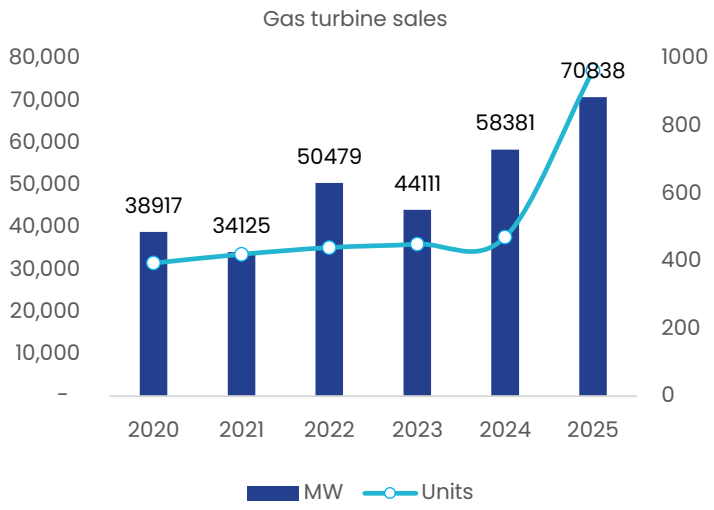
Story in charts

Electricity demand growth supporting turbine supply chains: Rising electricity consumption by 2050 will increase investments in power generation systems, supporting demand for turbine components



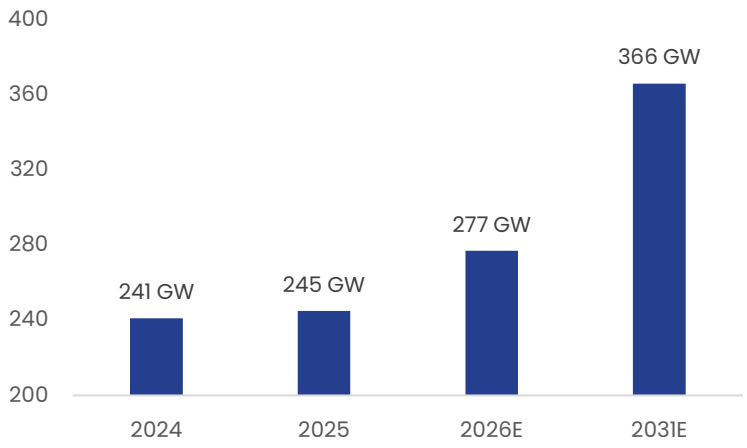
Source: IEA – World Energy Outlook

Gas turbine demand surge in 2025: Global gas turbine installations reached ~70.8 GW and 964 units in 2025. This supports demand for turbine components



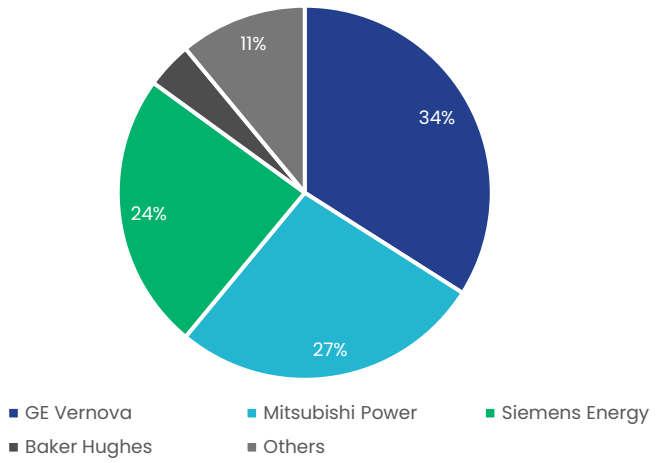
Source: Global Gas Turbine Market Forecast (2025 revision)

India's rising power demand: Peak electricity demand is projected to increase from 241 GW in 2024 to 366 GW by 2031, driving investments in power generation infrastructure and supporting long-term demand for turbine components



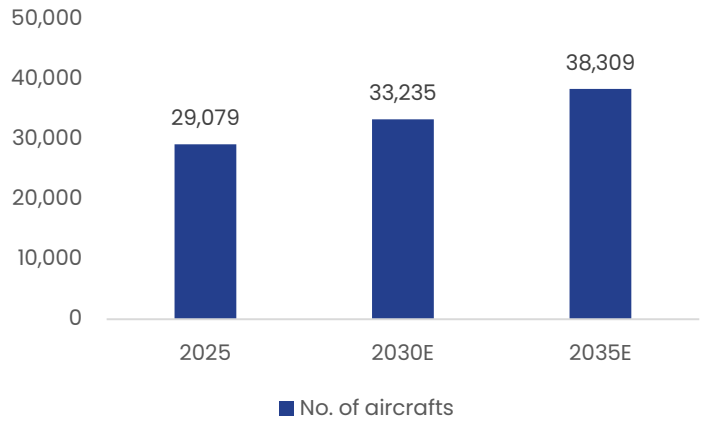
Source: PIB, Ministry of New and Renewable Energy

Highly concentrated OEM market: The global gas turbine market is dominated by a few OEMs such as GE Vernova, Mitsubishi Power and Siemens Energy, enabling suppliers to benefit from stable and long-term relationships



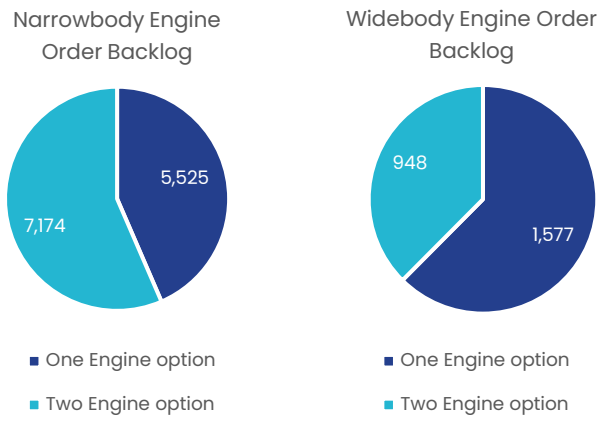
Source: Gas Turbine World market share analysis

Global airplane fleet forecast (2025–2035) implies ~32% growth, underscoring strong industry tailwinds



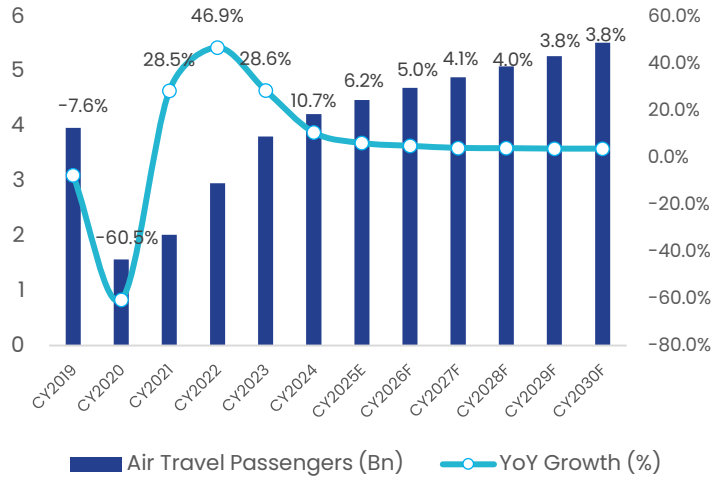
Source: Oliver Wyman analysis

Historic engine backlog across both types highlights a widening gap between supply and demand



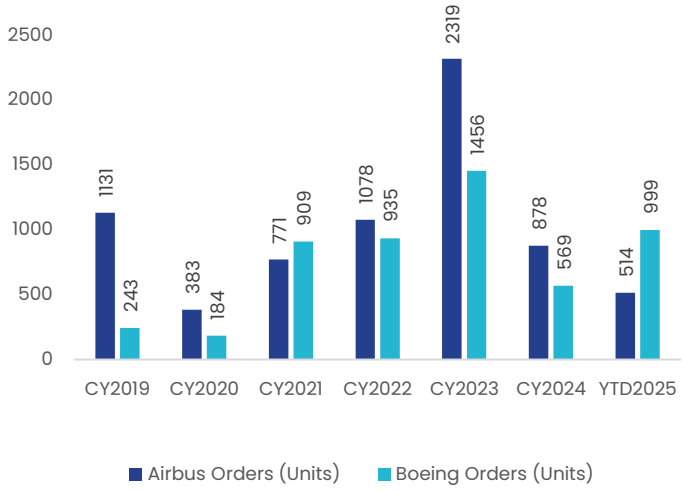
Source: Oliver Wyman analysis

Robust post-COVID recovery in air travel supports sustained momentum across the aero supply chain.



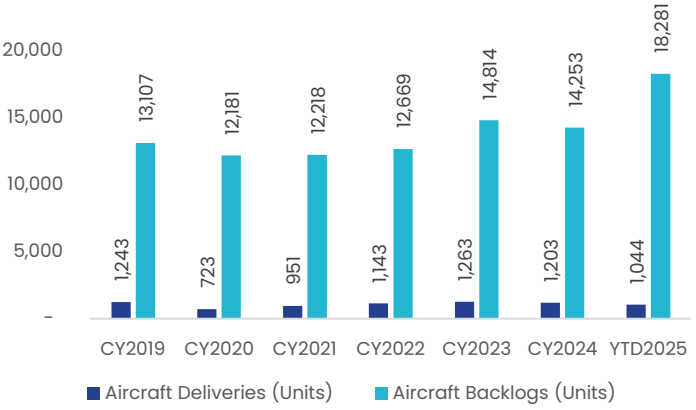
Source: IATA, ICAO, Frost and Sullivan Analysis

Total aircraft orders secured by airframers to date highlight durable sector visibility.



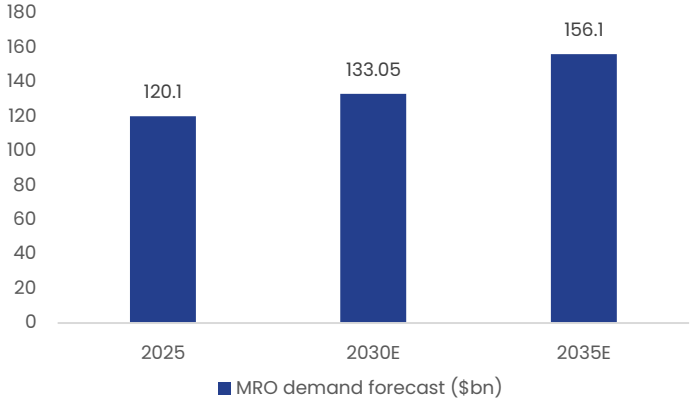
Source: Airbus and Boeing

Wide delivery–backlog gap at Airbus and Boeing underscores structural supply constraints



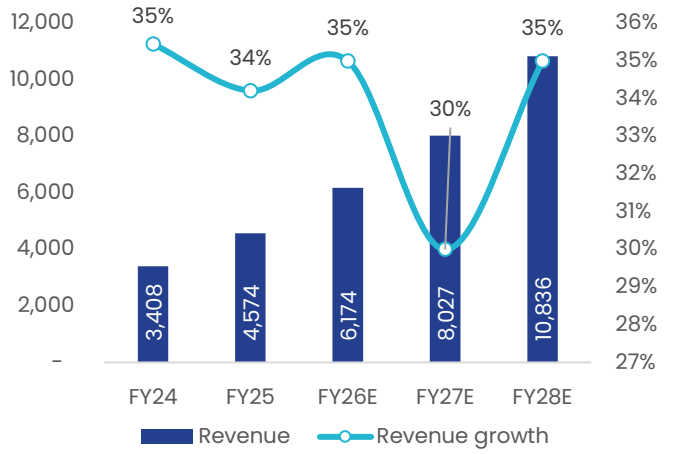
Source: Airbus and Boeing

An aging fleet that requires more servicing is contributing to a super cycle for the MRO market at a CAGR of 2.7% over 2025-2035



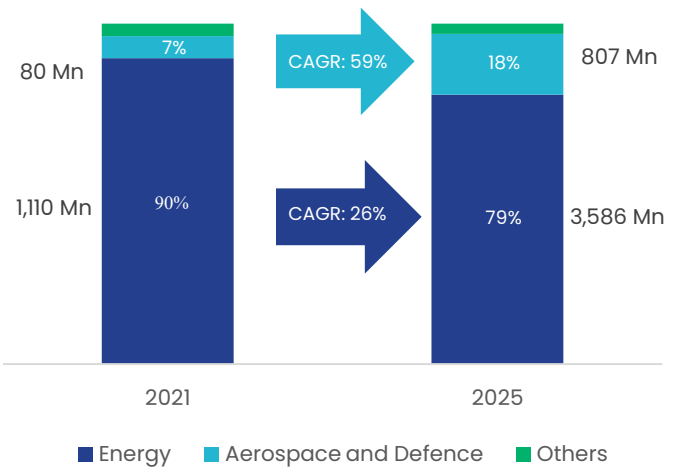
Source: Company, Ashika Institutional Research

Revenue expected to clock ~35% CAGR over FY24–FY28E, driven by strong order visibility and execution ramp-up



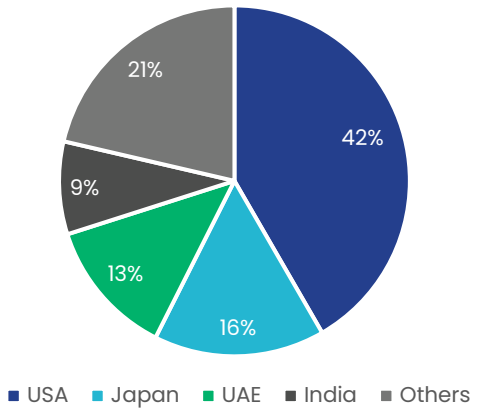
Source: DRHP, Company

Evolving segment mix with rising A&D contribution (18% in 2025), alongside steady energy growth at 26% CAGR



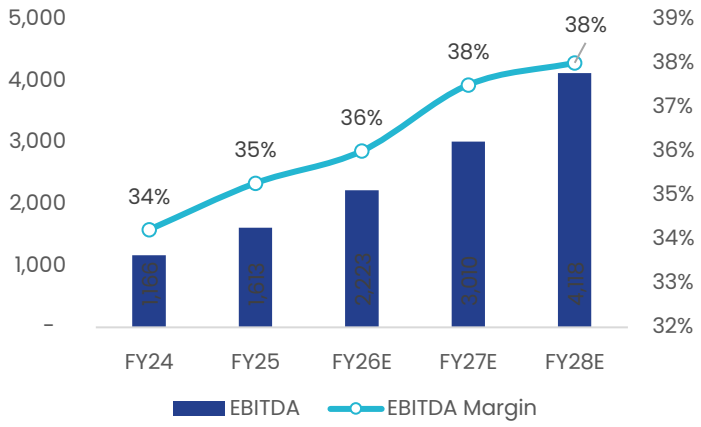
Source: AR, Company

Geographic revenue mix remains export-led, with the US and Japan dominating FY25 sales



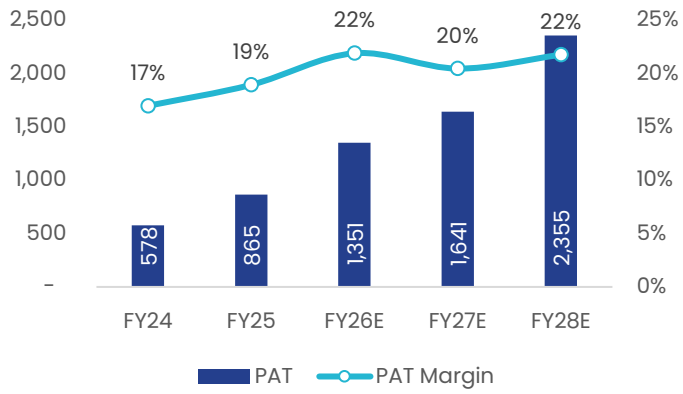
Source: AR, Company

EBITDA expected to scale materially over FY24–FY28E, supported by backward indigenisation & operating leverage



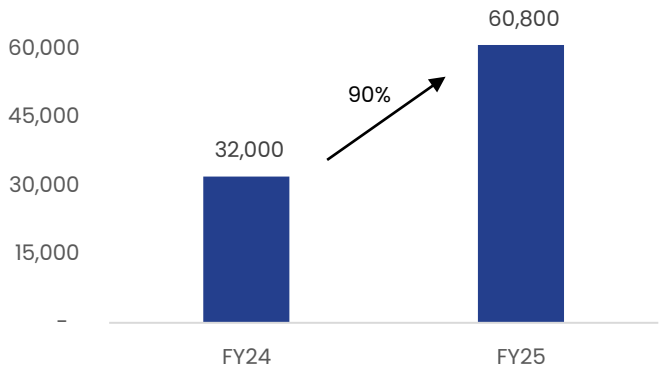
Source: AR, Company

Strong PAT expansion with margins sustaining in the ~20–22% range



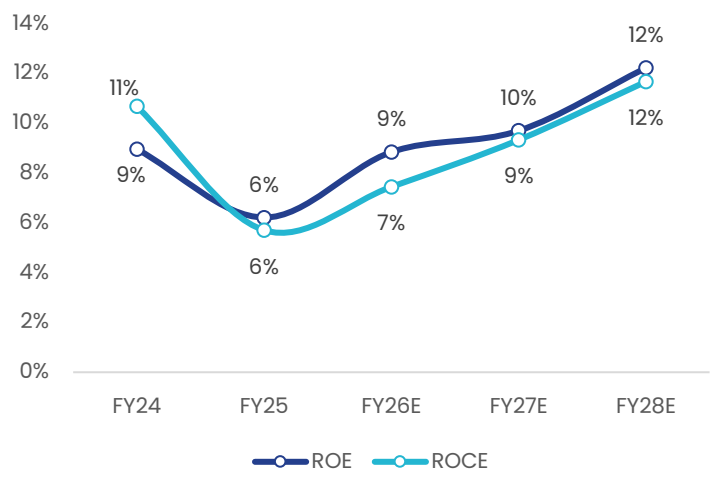
Source: AR, Company

Order book grew ~90% to ~Rs60,800mn, with client-dedicated facilities enhancing execution visibility



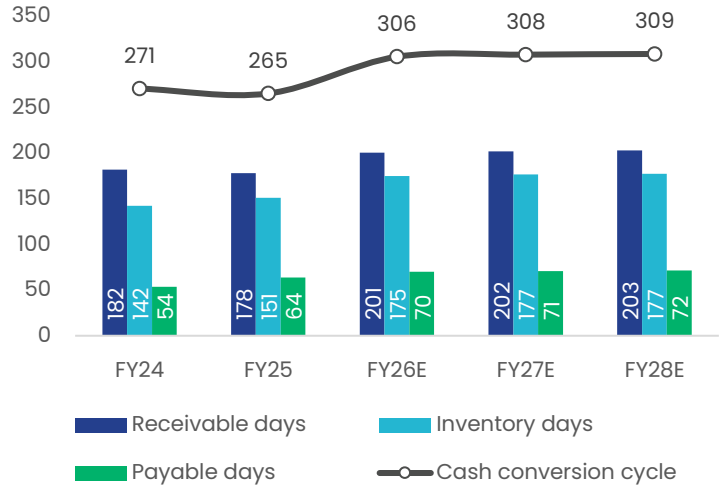
Source: AR, Company

ROE/ROCE expected to improve to ~12% by FY28E, driven by earnings ramp-up and better capital utilisation



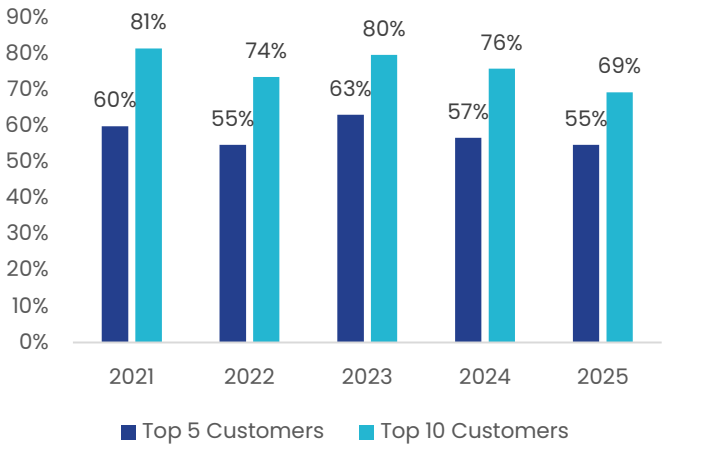
Source: DRHP, Company

Working capital remains elevated due to long qualification cycles and bulk inventory procurement during the approval phase



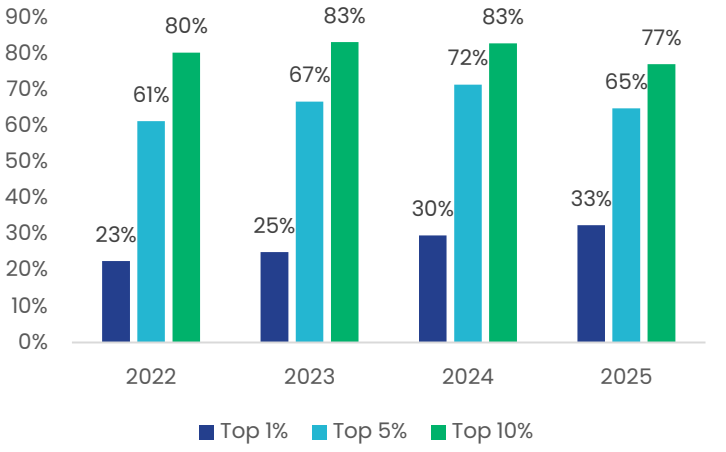
Source: AR, Company

Customer concentration

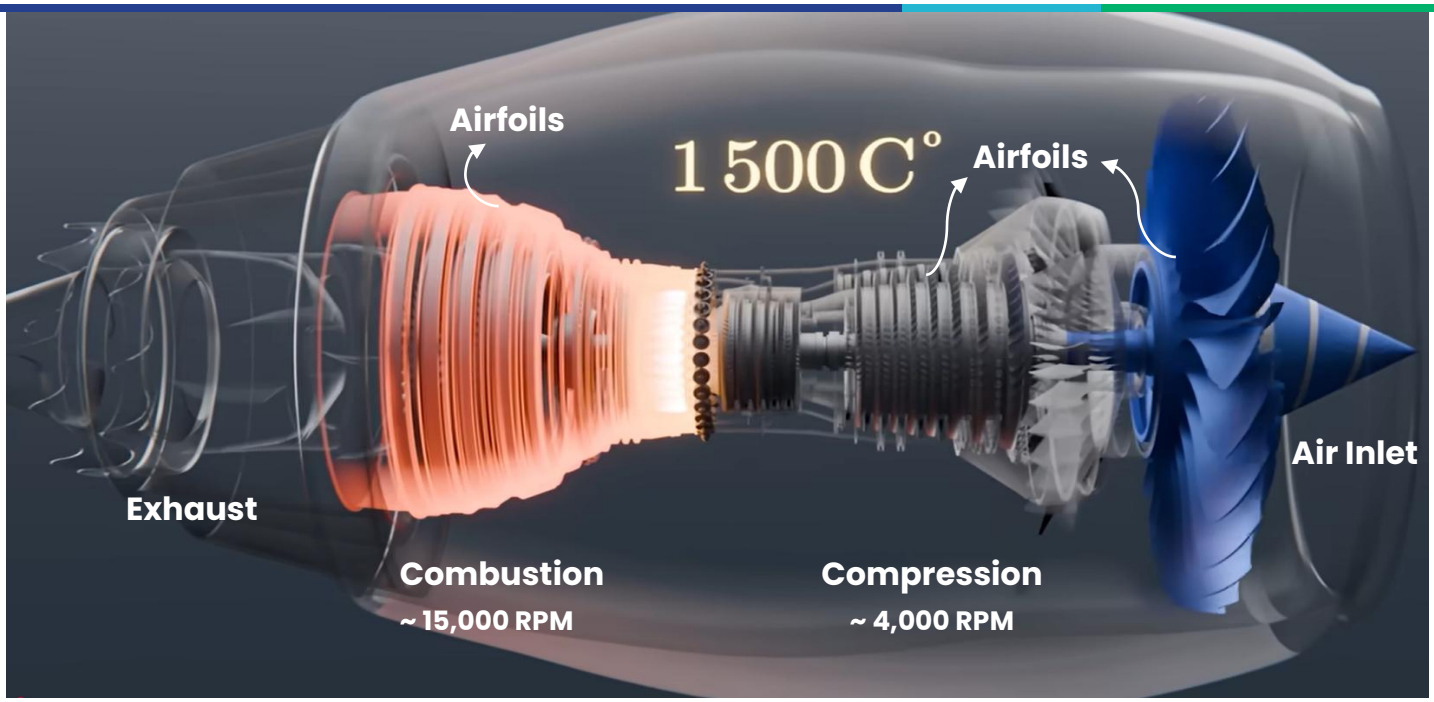


Source: RHP, Company

Supplier concentration



Source: RHP, Company



Azad currently **specialises in the compression section** of turbines, supplying high-precision airfoils and critical components. It is now **strategically expanding into the combustion section**. The company's long-term objective is to evolve from a component's supplier to a comprehensive, end-to-end turbine solutions provider, covering a larger portion of the turbine value chain and deepening its integration with global OEMs.

How does turbine work??

Air Intake → Air Compression → Increased velocity → More Air Compression → More Increased velocity → Addition of fuel → Explosion and Ignition → Engine Thrust

Basic principles of manufacturing process:

1. **Compressor**→ Takes air from outside turbine and increases pressure
2. **Combustion**→ Burns fuel and produces high pressure, high velocity gas
3. **Turbine**→ Extracts energy from gas coming from the combustor

How players differentiate each other in airfoil manufacturing:

Azad maintains that there is no standard way to make airfoil; it's a combination of **[MAN + MACHINE + ART]**

- Choice of input
- Choice of machinery
- Choice of fixture
- Choice of tools
- Choice of programming

"Make to Print" Manufacturing Process:

Raw Material

Customer approved sources

Testing in NABL Lab

Forging/Heat Treatment Pre Special Process

Equipment selection to achieve part tolerances

NADCAP approved equipment

Level 2/3 inspectors

Machining

Machine selection to achieve part tolerances

Insights for efficiency

Advanced statistical techniques

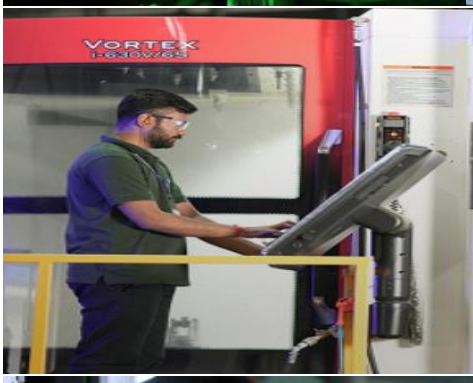
Post Special Process

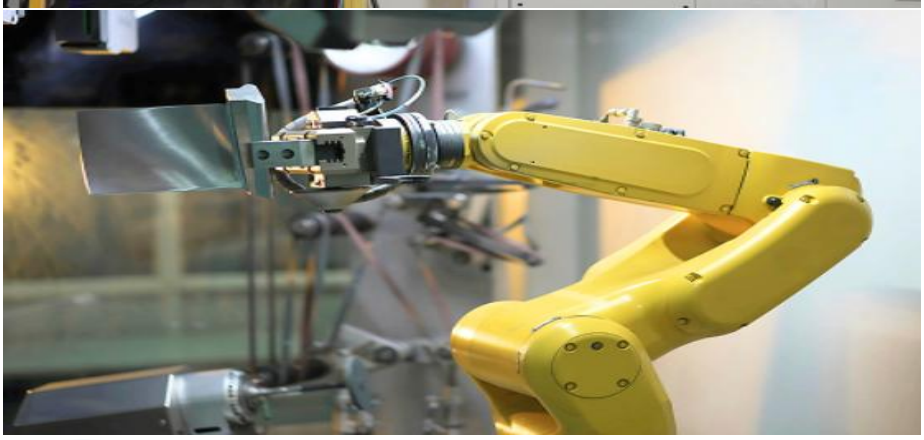
Equipment selection for required part process

NADCAP approved equipment

Level 2/3 inspectors

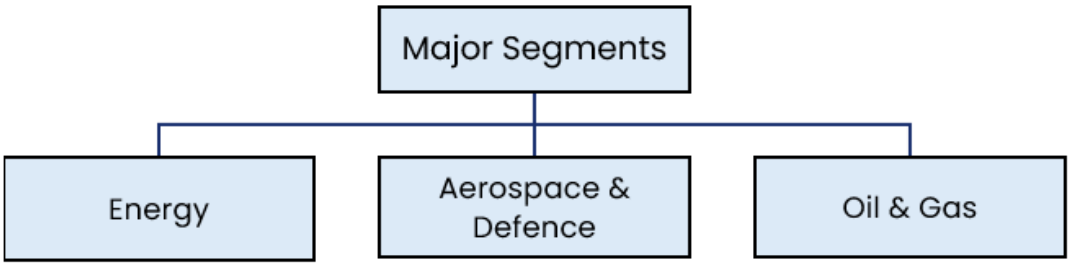
CNC Machine





About Azad:

Azad Engineering Limited was founded in 2008 by Mr. Rakesh Chopdar, and it is a precision engineering company serving the Aerospace & Defence, Power Generation and Oil & Gas industries worldwide. In an era shaped by energy transition and geopolitical realignment, the company has positioned itself as a strategic partner in industries that demand absolute precision, reliability and zero-defect execution. It operates where the "**margin for error" is non-existent**. From advanced gas turbines to next-generation aircraft, from nuclear platforms to defence propulsion systems, Azad's components are integral to technologies that power and protect the world. FY25 was more than a year of record numbers for Azad—it was a year of strategic consolidation. With marquee contracts, inauguration of dedicated facilities, a strengthened balance sheet and an expanded leadership team, Azad has laid a solid foundation for future. As it scales capacity and wallet share in segments where qualifications are now complete, the company is well positioned to capitalize on opportunities across Aerospace & Defence, Energy and Oil & Gas sectors.



Competition: Azad competes with manufacturers from the US, China, Europe, the Middle East and Japan. Its customers include global OEMs across Energy, Aerospace & Defence and Oil & Gas industries. Its components have been supplied to countries such as the US, China, Europe, the Middle East and Japan since inception. Accordingly, Azad is a key link in the global supply chain for OEMs in these three critical sectors.

Energy segment: In the Energy industry, Azad produces high-precision rotating and stationary 3D airfoils/blades, special machined parts and combustion component assemblies for land-based turbines with applications in industrial and energy plants using different fuel types such as nuclear, hydrogen, natural gas and thermal.

Aerospace & Defence segment:

In Aerospace & Defence, products include airfoils/blades and components for engines, auxiliary power units (APUs), hydraulics, actuating systems, flight controls, fuel and inerting sections of commercial & defence aircrafts and spacecrafts, besides other defence systems and various critical components for missiles. It has supplied critical components to major

aircraft manufacturers for models such as the B737, B737 Max, B747, B777, B777X, A320, A350, A355, A350 XWB and Gulfstream G550.

Oil & Gas segment:

For the Oil & Gas industry, it manufactures components for drilling rigs such as drill bits and other critical parts that are used in drilling equipment and are part of exploration and production phase.

Investment thesis:

Our investment thesis is based on the following premises:

1. **High entry barriers;** *Driven by mission-critical products and deep OEM integration*
2. **Diversified end-industry profile**
3. **Strong order visibility;** *Supported by capacity expansion and OEM-aligned capex*
4. **Other potential revenue streams;** *Entry into combustion and GTRE*

Azad's negotiated price contract leads to sustainable EBITDA margins:

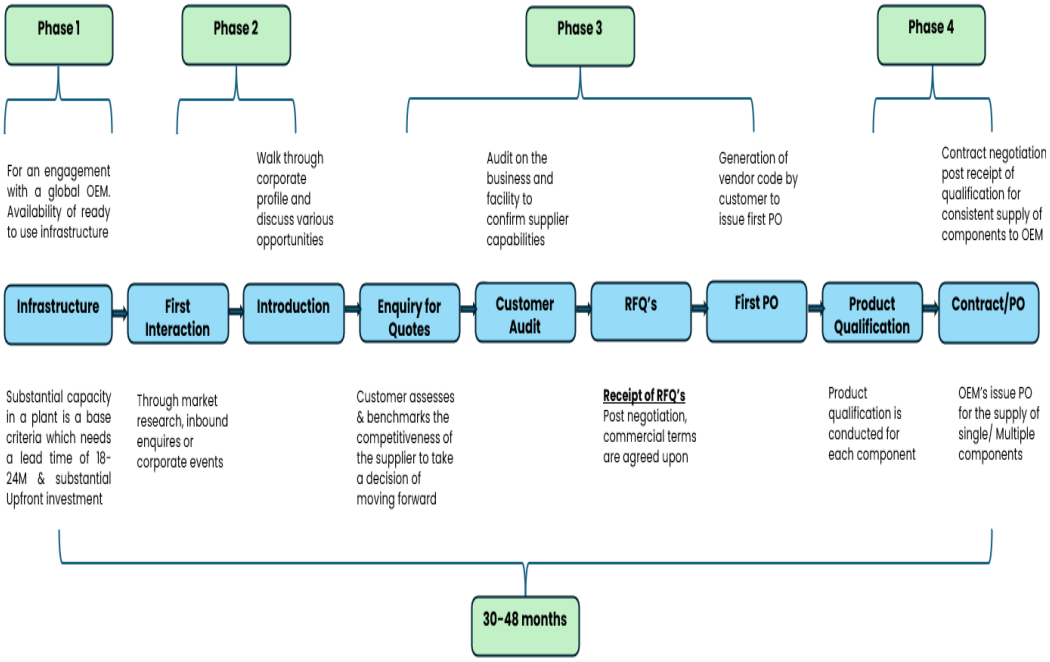
RM Price Fluctuation
+/- 4-5% → Pass-through to OEM
>5% → Absorbed by supplier

1. **High entry barriers;** *Driven by mission-critical products and deep OEM integration*

Azad operates in a niche segment, manufacturing mission-critical and life-critical components **with zero-defect requirements**, where precision, reliability and process expertise are non-negotiable. With a portfolio of **1,700+ qualified parts and 45+ specialised manufacturing processes**, the company has built capabilities that are difficult to replicate. Entry barriers are further reinforced by **stringent, multi-year qualification cycles**—where each component and facility must undergo rigorous audits, testing and approvals—making vendor replacement extremely challenging once qualified.

This has enabled Azad to establish **long-term strategic partnerships with global OEMs** such as GE Vernova, Mitsubishi Heavy Industries, Rolls-Royce, Honeywell and Siemens Energy, with client relationships often exceeding a decade and supported by multi-year contracts. Its **Rs60bn+ order book** reflects strong customer confidence and embeds the company deeply within global supply chains. The combination of high qualification barriers, zero-defect execution and long-standing OEM relationships results in **strong customer stickiness, high repeat business and durable competitive advantage.**

Qualification Process:



2. **Diversified end-industry profile and its growing TAM:**

In Aerospace & Defence, the global components TAM is projected to grow from ~Rs990bn in 2022 to **~Rs1,530bn by 2027** (~9% CAGR), supported by strong aviation demand, rising RPKs and a record backlog of >17,000 aircraft (led by Airbus and Boeing). Azad is well positioned with mission-critical components across major platforms, including Boeing 737, Airbus A320/A350 and Gulfstream G550; the segment delivered Rs807mn revenue in FY25 (17.9% mix), growing 84.1% on the back of completed qualification cycles and execution strength.

In Energy and Oil & Gas, the turbine components market (~Rs280bn in 2022; part of a broader Rs1,280bn TAM expected to reach **Rs1,810bn by 2027**) remains Azad's core segment, contributing Rs3.6bn (79% of FY25 revenue), up 26%. The company supplies compressor airfoils, SMP/CMP components and combustion parts to OEMs and controls ~75% of the global gas turbine market while also expanding its Oil & Gas portfolio across upstream and midstream applications.

The global airfoil and turbine blade ecosystem:

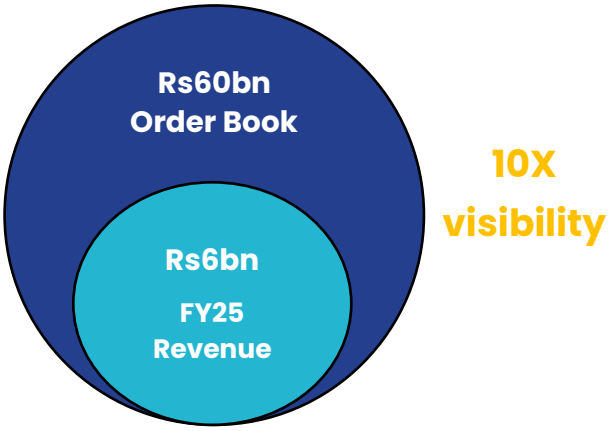
Segment	2024 size	2032-35E size	CAGR
Aircraft engine blade (total, all types)	$13.5-15.6B	$21-28B	5.5-6.9%
- Airfoil shape segment	~$5.0B	~$9.0B	~6%
- Nickel-based superalloy segment	~$6.0B	~$10.0B	~5%
- Turbine blade sub-segment	~$4-5B	~$7-8B	~6%
Aircraft engine casting (investment casting)	~$8-10B	~$15-18B	~6%
Industrial gas turbine components	~$5-6B	~$8-9B	~4-5%
Steam turbine market (equipment)	$17-19B	$21-23B	~2.5%
Steam turbine services/MRO	$19.5B	$31.8B	~5%

Within this, the addressable market for Azad spans multiple sub-segments: precision-machined airfoils for energy turbines (gas, steam, nuclear), aerospace engine components (blades, vanes, actuators) and defence propulsion parts. The airfoil-specific segment (by blade shape) was ~USD 5 billion in 2024 and is expected to reach USD 9 billion by 2035. The nickel-based superalloy segment alone was valued at USD 6 billion in 2024

Azad's FY25 revenue of Rs5.58bn (~USD 65 million) represents less than 0.5% of the global aircraft engine blade market. Even within the narrower precision-machined airfoil niche (estimated at USD 3-4 billion globally), Azad's share is ~1.5-2%. This underscores the enormous runway for growth
as Azad scales production and adds new OEM qualifications.

3. **Strong order visibility supported by capex and OEM-aligned capex**

Azad closed FY25 with a robust order book exceeding Rs60bn, comprising largely long-term contracts that provide strong multi-year revenue visibility and reflect its growing embeddedness in global Aerospace, Energy and Defence supply chains. Recent order wins across core verticals highlight its ability to consistently secure high-value programs, including marquee government and indigenisation opportunities in India. To support scalable execution, the company has commissioned client-dedicated lean manufacturing facilities for Mitsubishi Heavy and GE Vernova under its Phase-1 expansion at Hyderabad, designed to global specifications and aligned with customer processes. These investments enhance operational efficiency, improve turnaround times and strengthen Azad's ability to scale alongside long-term OEM demand.



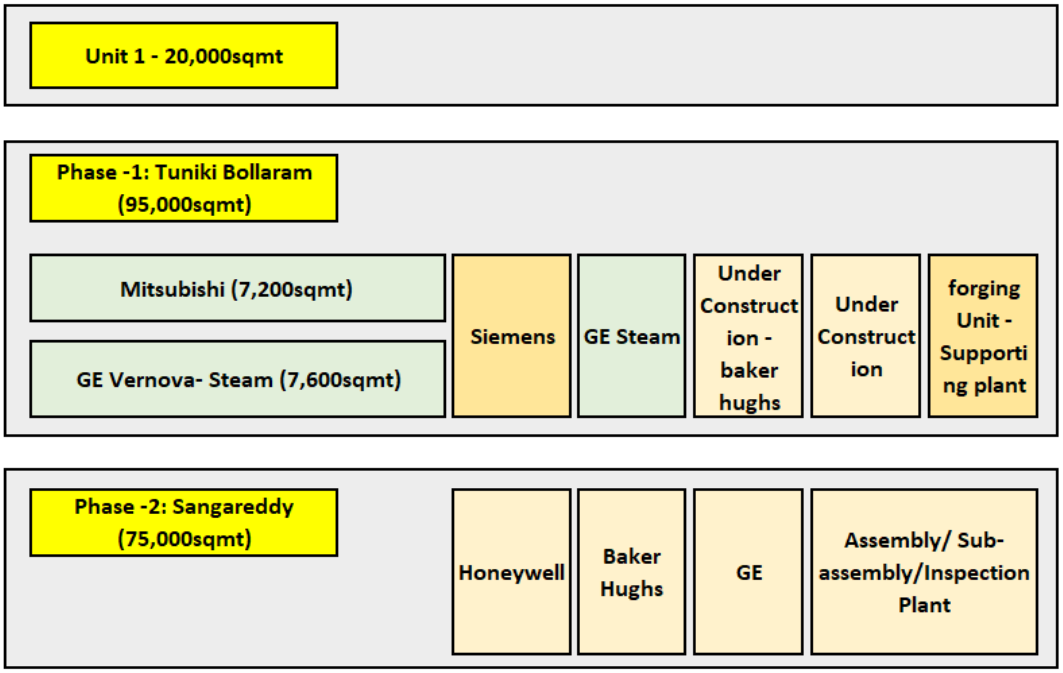
Few key orders won over the years:

Date	Customer Name	Contract Amount (Rs. Mn)	Time
29-01-2024	Rolls-Royce	-	7 years
06-03-2024	Baker Hughes	-	5 +3 Years
13-03-2024	GE Vernova	3,115	7 years
23-05-2024	GTRE	-	
11-07-2024	Siemens Energy Global GmbH & Co	-	5 Years
24-10-2024	Honeywell Aerospace ISC, USA	1,424	-
03-11-2025	Mitsubishi Heavy Industries Limited, Japan (MHI). **(Facility inauguration on 12.03.2025)**	7,000	5 Years
12-11-2024	Arabelle Solutions France	3,400	5 years
15-01-2025	GE VERNOVA INTERNATIONAL LLC, USA (Gas & Thermal Turbine) (**Facility inauguration on 29.04.2025)**	960	6 years
27-01-2025	Bharat Heavy Electrical Limited (BHEL)	-	-
29-01-2025	Siemens Energy Global GmbH & Co. KG **(Facility inauguration on 18.09.2025)**	8,110	6 years
05-02-2025	Rolls- Royce PLC	-	Contract
04-05-2025	GE Vernova (Steam Turbine)	4,525	6 Years
19-05-2025	Nuovo Pignone srl (Baker Hughes) - Addition to Contract on 06.03.24	-	5 Years
26-09-2025	Mitsubishi Heavy Industries Limited	6,510	5 Years

Source: Company, Ashika Institutional Research

Rs 2bn
Est. Revenue/Factory
x
9 facilities
New & Planned
=
Rs 18bn
Potential revenue visibility
+
Rs. 4.5bn
Existing facility Revenue
=
Rs. 22.5bn
Projected Revenue (with existing CAPEX plans)

OEM-aligned capex: Capacity is about to reach 10x with completion of phase-1:



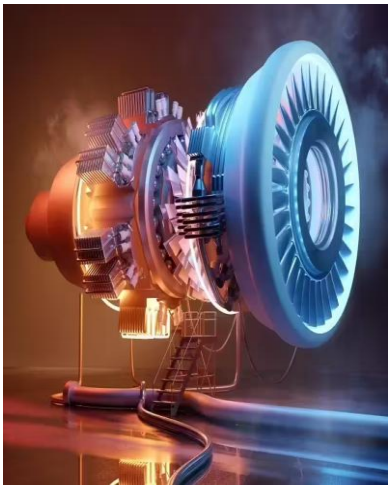
Source: Company, Ashika Institutional Research

4. Other potential revenue streams

a. <u>Expanding from compressor airfoils into combustion components</u>

Azad is expanding beyond compressor airfoils into combustion components, which represent a more complex and higher-value part of the turbine. This transition allows the company to increase its wallet share per turbine platform and deepen its integration with global OEMs. Over time, it strengthens Azad's positioning as a more critical supplier across the turbine value chain.

b. <u>Advanced turbo gas generator engine with GTRE</u>



Azad has secured a prestigious nation's pride contract from GTRE [Gas Turbine Research Establishment), one of the pioneering Research & Development organizations under the DRDO [Defence Research and Development Organisation] and the Ministry of Defence, Government of India, for the end-to-end manufacturing, assembly and integration of a completely assembled Advanced Turbo Gas Generator Engine. This engine is a critical development for India and will be utilised in various defence applications. This milestone marks Azad's entry into propulsion systems, showcasing its growing presence and capabilities in the field.

Azad will play a vital role as a single source Industry Partner (IP) to GTRE for the success of this mission.

This vertical integration and enhancement in capability enables Azad to offer end-to-end manufacturing for a complete, assembled Advanced Turbo Engine, creating massive opportunity to build capability for a complete propulsion systems solution for the nation and global defence equipments.

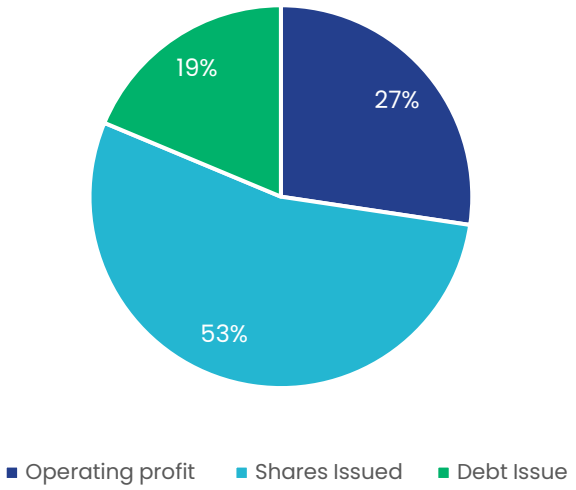
Azad will start delivering first batch of fully integrated turbo engines by early 2026.

Azad's GTRE engagement is strategically important for defence aero-engine qualification, with limited near-term revenue (Rs50–150mn), but meaningful long-term optionality (Rs300–1000mn+), contingent upon indigenous engine serialisation.

Fund flow analysis:

Cumulative cash inflow over the last five years

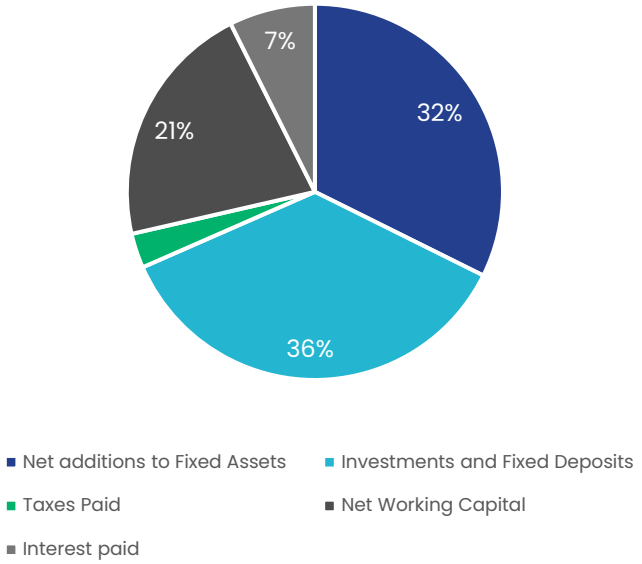
Over the last five years, the company's cash inflows have been supported by a combination of equity issuance (53%), operating profit (27%) and debt funding (19%), providing capital to fund its expansion initiatives



Source: Company, Ashika Institutional Research

Cumulative cash outflow over the last five years

On the outflow side, most funds have been deployed toward investments, fixed deposits (36%), and additions to fixed assets (32%), reflecting disciplined capital allocation toward growth and capacity expansion while supporting working capital needs



Source: Company, Ashika Institutional Research

Plant visit takeaways:

1. **Entrepreneurial origins; execution-led scale-up:** Promoter Mr Rakesh Chopdar, a school dropout, began with his family's small fasteners business (~Rs. 1mn revenue). Early exposure to OEM requirements led him to replicate a rejected precision component without formal infrastructure—marking Azad's entry into high-precision manufacturing. Initial job-shop orders (incl. Saudi clients) scaled into airfoils manufacturing followed by steady capability build-out.
2. **Airfoils manufacturing remains capability-driven:** There is no standardized production pathway; outcomes depend on integrated execution across input material selection, machinery, tooling, fixtures and programming. Management views airfoils manufacturing as an execution-led discipline **("man + machine + methodology")**, forming a key competitive moat.
3. **Positioned in compressor airfoils; moving up the value chain:** Azad primarily supplies compressor-stage airfoils for industrial gas turbines, where multi-stage compressors (17–22+ stages) require highly customized blades across dimensions and tolerances. OEM qualification typically begins with lower-stage parts and scales with consistency. Management highlighted a roadmap to expand into combustion components (higher heat tolerance), implying deeper turbine integration over time.
4. **Sticky qualification-led business model:** Entry barriers remain high given the OEM-led approval processes involving RFQs, FAI validation, RM/coating approvals, batch procurement requirements, and consistent execution across price, lead time and quality parameters. This supports long product cycles and strong client stickiness once qualified.
5. **Build-to-print model with pricing protection:** Most contracts follow fixed-price structures with limited RM volatility absorption (~4–5% threshold); beyond this, costs are typically passed through, preserving margin stability over cycles.
6. **Demand tailwinds linked to outlook for power demand:** Management highlighted stronger-than-expected growth in global electricity demand (already ~3% vs earlier 2–2.5% projections), supporting OEM order flows (GE Vernona, Mitsubishi, Siemens Energy) and downstream suppliers like Azad.
7. **Capacity-led growth visibility:** Current facilities can support ~Rs.4.5bn revenue. With Phase-1 expansion (funded via ~Rs7bn QIP), total potential revenue capacity rises to ~Rs20bn. Blended asset turn is ~1.8x (Energy ~1.7x; Aerospace ~2.5x).
8. **Strategic investments to deepen ecosystem:** Azad has made small strategic acquisitions (incl. Azad VTC) to strengthen vertical integration and execution capability.
9. **WC structurally elevated but improving:** Working capital intensity reflects long qualification cycles and batch RM procurement from approved vendors; management targets normalization to ~140–160 days by FY27.

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Peer Comparison:

Company	Revenue (Relevant Segment)	Key Products	Key OEM Customers	Competitive Position / Moat
Azad Engineering (India)	Rs 5.58bn / ~$65M (FY25, TTM)	3D rotating airfoils, turbine blades, precision machined components	GE, Honeywell, Siemens, MHI, Eaton	1,700+ qualified parts; zero PPM track record; GTRE single-source AGTE partner; 45+ processes
Howmet Aerospace (US)	$3.7B (Engine Prods, FY24)	Aero airfoils (SC, DS, EQ), IGT, structural castings	GE, P&W, RR, Safran	World #1 in SC airfoils; vertically integrated alloy-to-coating
PCC Airfoils / Precision Castparts (US)	~$2.5B (est. ICP segment)	Airfoil investment castings, structural castings, forgings	GE, P&W, RR, Boeing	Berkshire Hathaway owned; #1 in structural castings; every jet engine program
Chromalloy (US)	~$800M-1B (est.)	Airfoil coatings, castings, repairs; IGT components	GE, Siemens, P&W	Veritas Capital owned; strong in MRO/aftermarket; Advanced Airfoil Components (sold to Siemens 2022)
Doncasters Group (UK)	~$500-700M (est.)	Turbine airfoils, compressor airfoils, rings & casings	GE, RR, Siemens, Westinghouse	Dubai Intl Capital owned; acquired Uni-Pol (2022) for superalloy castings
Consolidated Precision Products (US)	~$400-600M (est.)	Investment castings for aero & IGT; acquired Selmet Inc.	GE, P&W, RR	Warburg Pincus backed; strong in structural and airfoil castings
Hitchiner Manufacturing (US)	~$300-400M (est.)	Investment castings; counter-gravity casting specialist	GE, P&W, Honeywell	Pioneer in counter-gravity casting; strong in smaller complex airfoils
IHI Corporation (Japan)	~$2B (Aero Engine segment)	Aero engine components, fan blades, LPT modules	P&W (RRSP partner), GE	Revenue share partner on PW1100G (A320neo); makes full LPT modules
Impro Precision Industries (HK/China)	~$250-350M (est.)	Investment castings, machining for aero & IGT	GE, Siemens, Honeywell	Listed in HK; expanding aerospace share; competitive on cost
TURBOCAM International (US)	~$150-250M (est.)	5-axis machined flow-path components: blades, vanes, blisks, impellers	GE, P&W, RR, Honeywell	150+ 5-axis machines; 4 countries; AS9100D; Nadcap coatings
Hi-Tek Manufacturing (US)	~$50-100M (est.)	Precision turbine blade machining from superalloys	GE, Siemens, Solar Turbines	125,000 sq ft facility; specialises in complex cooling hole drilling
Moeller Aerospace (US)	~$50-80M (est.)	Turbine airfoil machining: blades, vanes, nozzles, seals	GE, P&W, RR	50%+ volume is blade/vane machining; Click-Loc tech; 3 patents filed
PTC Industries / Aerolloy (India)	~Rs 7bn / ~$85M (FY25E)	SC/DS castings, titanium castings, post-cast blade machining	GTRE/DRDO, Safran, HAL	First Indian co. for SC casting + post-cast ops; acquired Trac Precision (UK); LEAP engine supplier to Safran
Godrej Aerospace (India)	~Rs 4-5bn (est.)	Engine modules, turbine blade brazing, combustors, KDE production	GTRE, ISRO, Honeywell, RR, Boeing	Delivered first serial KDE engine; Kaveri turbine blade maker; ISRO liquid engine sole fabricator
Bharat Forge (India)	~Rs 18bn (Aerospace BU est.)	Forgings, compressor discs, small turbojet engines (40-400 kgf)	GE, RR, Boeing, DRDO	NADCAP certified; developing ab-initio gas turbine engines; Zorya Mashproekt acquisition

Competitive Deep-Dive:

Mapping of the six companies globally that operate **as independent, specialist precision machine manufacturers for turbine airfoils, blades, vanes and hot gas path components** — the exact business model as Azad Engineering. These are **build-to-print suppliers** serving OEMs like GE, Pratt & Whitney, Rolls-Royce, Siemens Energy and Honeywell.

Company	Revenue	Employees	Rev/Employee	Location	Ownership	Key OEMs
TURBOCAM International	$158M	1,000+	$158K	US, India, UK, Romania	Private (Family)	P&W, GE, Cummins, ISRO
Moeller Aerospace	$146M	~400	$365K	US (Michigan, 2 plants)	PE (AE Industrial)	GE, P&W, RR
Azad Engineering	**$65M**	**~1,500**	**$43K**	**India (Hyderabad)**	**Public (NSE/BSE)**	**GE, Honeywell, Siemens, MHI**
Tribus Aerospace	$35M	~250	$140K	US (MI, OH, CA)	PE (Shorehill)	Aero & Defence OEMs
C*Blade S.p.A.	$11M	50–250	$44–220K	Italy (Maniago)	PE (Ethica Global)	Major power gen OEMs
Hi-Tek Manufacturing	$10M	~150	$67K	US (Mason, OH)	Private	All major turbine OEMs

TURBOCAM is Azad's single most important benchmark. Both companies are precision 5-axis machining houses serving the same OEMs with similar products. TURBOCAM has 2.4x Azad's revenue but a fundamentally different cost structure: $158K revenue per employee (US/EU wages) vs. Azad's $43K (India wages). TURBOCAM's India operation in Goa (31 machines, ~120 employees, ~$6M revenue) is small relative to Azad's Hyderabad complex. Critically, TURBOCAM is private and family-owned, limiting its access to growth capital. Azad's public listing on NSE/BSE gives it a structural fundraising advantage.

Competitive risk to Azad: TURBOCAM's Goa facility could be scaled up to compete directly for India-cost-advantage work. Their PW GTF vane plant shows they can win dedicated production contracts on next-gen programs. However, TURBOCAM's focus is broader (turbochargers are a large revenue share), diluting their competitive intensity against Azad in the energy airfoil niche.

Capability Comparison Matrix:

The following matrix compares the technical capabilities of all six "Tier A" peers across the key processes required for turbine blade and airfoil machining. A checkmark (✓) indicates confirmed capability; a cross (✗) indicates no evidence of capability.

Capability	TURBOCAM	Moeller	Azad	Tribus	C*Blade	Hi-Tek
5-axis CNC milling	✓ (150+ machines)	✓	**✓ (45+ processes)**	✓	✓	✓
EDM (sinker/wire/fast-hole)	✓ (ECM too)	✓	✓	✓	✗	✓
Creep-feed grinding	✓	✓	✓	✓	✗	✓
Laser drilling/processing	Limited	✗	✓	✗	✗	✓
Vacuum heat treatment	✗	✗	✓	✗	✓	✓
Brazing	✗	✗	✓	✗	✗	✓
Forging capability	✗	✗	**✓ (precision forging)**	✗	✓ (drop forging)	✗
Titanium aluminide machining	✗	✓ (World leader)	**Limited**	✗	✗	✗
Blisk machining (IBRs)	✓ (Core competency)	Limited	**Limited**	✗	✗	✗
Coatings (TBC/PVD)	✓ (Nadcap)	✗	✓	✗	✗	✗
NDT/FPI in-house	✓	✓	✓	✓	✓	✓
AS9100D certified	✓	✓	✓	✓	✓	In progress
Nadcap accreditation	✓ (Coatings)	✗	✓	✗	✗	✗
India manufacturing	✓ (Goa, 31 machines)	✗	**✓ (Hyderabad HQ)**	✗	✗	✗
Aero engine parts	✓	✓	✓	✓	✗ (power gen only)	✓
IGT/power gen parts	✓	✓	**✓ (largest segment)**	✓	✓ (core focus)	✓
Turbocharger parts	✓ (large volume)	✗	✗	✗	✗	✗
Defence/DRDO contracts	✗	✗	**✓ (GTRE single-source)**	✓	✗	✗
OEM qualification depth	Thousands of designs	Extensive (since 1949)	**1,700+ qualified parts**	Moderate	60+ years energy	40+ years

Key observations: Azad has the broadest capability set among Tier A peers — spanning forging, 5-axis machining, EDM, laser, heat treatment, brazing and coatings, all of which is done in-house. TURBOCAM leads in blisk/IBR machining and ECM. Moeller is the undisputed leader in titanium aluminide. Only Azad and TURBOCAM have India manufacturing operations. Only Azad has active defence/DRDO production contracts.

Azad Engineering vs Howmet Aerospace

Howmet Aerospace (HWM US) is the closest global peer to Azad Engineering; both companies manufacture high-precision rotating components (blades, vanes, housings & airfoils) for aero-engines and industrial gas turbines, supply to the same OEM customers (GE, Siemens, Honeywell, Rolls-Royce & Mitsubishi), and derive their competitive moat from multi-year OEM qualification cycles that create near-impenetrable entry barriers.

Howmet's trajectory over 2020–2025 — revenue growing from $4.3bn to $8.3bn, EBITDA margin expanding from 22% to 29%, PE re-rating from 23x to 65x, and stock delivering ~400% returns — is the most credible template for Azad's own journey. Azad is, in essence, a **'Baby Howmet'** operating from an Indian cost base, with structurally higher margins, materially faster near-term growth and a much longer runway.

Azad CMP: Rs.1,422 | Market cap: ~Rs91.87bn (~US$1.0bn) | Howmet market cap: ~US$74bn | Azad = ~1.4% of Howmet's market cap today

1. Business comparability — Why Howmet is the right peer

Dimension	Azad Engineering	Howmet Aerospace
Core Products	Precision airfoils, turbine blades & vanes, compressor blades, rotating components, structural housings	Engine blades & vanes, structural castings, fasteners, forged wheels — all high-tolerance precision parts
End Markets	Aeroengines (~17% A&D mix, growing) | Industrial gas turbines (~65%) | Oil & Gas (~5%, growing)	Commercial aerospace (~60%) | Defense (~15%) | Industrial gas turbines (~10%) | Oil & Gas, automotive (~15%)
Key OEM Customers	GE Vernova, Siemens Energy, Mitsubishi Heavy Ind., Honeywell, Rolls-Royce, Arabelle, Baker Hughes	GE Aerospace, Airbus, Boeing, Rolls-Royce, Safran, Honeywell, RTX, Pratt & Whitney
Manufacturing Process	5-axis CNC precision machining; AS9100D certified; multi-year OEM approval cycles	Aerospace castings, forgings, machining; OEM-qualified with sole-source positions; 5–7-year qualification cycles
Competitive Moat	3–5-year OEM qualification cycles; proprietary machining capabilities; deep application-specific expertise	Sole-source positions on ~70% of parts; OEM qualification cycles create near-permanent lock-in
Geography	India (Hyderabad); ~80% revenues from exports; expanding production capacity aggressively	US, UK, France, Czech Republic; global manufacturing footprint
Stage of Cycle	High-growth: new plants commissioning; order book Rs6,000 Cr; 25–35% revenue CAGR guidance	Mature compounder: deleveraging, FCF generation, margin expansion; ~10% revenue growth

2. Revenue & EBITDA — Growth Trajectory

Metric	FY21	FY22	FY23	FY24	FY25	FY26E	FY27E
AZAD ENGINEERING (Rs Crores)							
Revenue (Rs Cr)	**122.7**	**194.5**	**251.6**	**340.8**	**453.0**	**617.0**	**802.1**
Revenue Growth	—	58.4%	29.3%	35.5%	32.9%	34.9%	30.0%
EBITDA (Rs Cr)	**28.6**	**62.3**	**72.3**	**116.6**	**161.3**	**222.1**	**296.8**
EBITDA Margin	23.3%	32.0%	28.7%	34.2%	36.3%	36.0%	37.0%
EBITDA Growth	—	117.8%	16.1%	61.3%	38.3%	37.7%	33.6%
PAT (Rs Cr)	**—**	**—**	**8.1**	**57.8**	**88.5**	**131.0**	**140.7**
PAT Margin	—	—	3.2%	17.0%	19.6%	21.2%	17.5%
4Y Revenue CAGR (FY21–25): ~38.6% | 4Y EBITDA CAGR (FY21–25): ~54.1%							
HOWMET AEROSPACE (USD)							
Revenue	**—**	**$5.67B**	**$6.64B**	**$7.43B**	**$8.25B**	**~$8.9B**	**~$9.5B**
Revenue Growth	—	—	17.1%	11.9%	11.0%	8.0%	7.0%
Adj. EBITDA	**—**	**$1.27B**	**$1.51B**	**$1.92B**	**$2.1B**	**~$2.3B**	**~$2.5B**
EBITDA Margin	—	22.4%	22.7%	25.8%	25.5%	25.8%	26.3%
Adj. EPS	**—**	**$1.40**	**$1.84**	**$2.81**	**$3.42**	**$3.70**	**—**
3Y Revenue CAGR (CY22–24): ~13.3% | 3Y EBITDA CAGR (CY22–24): ~18.3% | 3Y EPS CAGR: ~35%							

Revenue: Azad's 4-year revenue CAGR of ~39% (FY21–FY25) outpaces Howmet's ~13% over the same period — reflecting Azad's early-stage hyper-growth from a small base. Both operate in the same secular demand upcycle (global aero engine demand + industrial gas turbines), meaning Azad's trajectory is structurally validated by what its global peer is experiencing.

EBITDA: Azad's EBITDA margin has expanded from 23% (FY21) to 36% (FY25) and is guided to reach 36–38% by FY26–FY28. Critically, Azad's margins already exceed Howmet's 25–26% levels on a much smaller revenue base — a function of India's labour and power cost advantage. This is not a one-time phenomenon: at scale, Azad should sustain 35–38% EBITDA margin vs Howmet's ~29–30% at US$8bn+ revenue.

3. Valuation Multiples — P/E, P/BV, P/CF

Valuation Metric	FY22	FY23	FY24	FY25
PRICE-TO-EARNINGS (P/E)				
Azad EPS (Rs)	—	4.9	9.8	13.4
Azad PE @ CMP Rs1,422	—	—	—	106x
Howmet EPS (USD)	—	$1.40	$1.84	$2.81
Howmet PE (trailing)	—	—	28x	46x
PRICE-TO-BOOK (P/BV)				
Azad Book Value/Share (Rs)	—	124	109	216
Azad P/BV @ CMP Rs1,422	—	13x	7x	6x
Howmet Book Value/Share	—	$5.85	$6.79	$8.83
Howmet P/BV	—	7.5x	8.3x	10.0x
PRICE-TO-CASH FLOW (P/FCF)				
Azad Operating CF (Rs Cr)	-	(10)	(7)	54
Azad FCF (Rs Cr)		(111)	(62)	(864)
Howmet FCF ($B)	—	—	$0.68	$0.98
Howmet P/FCF	—	—	—	47x

P/E: At the CMP of Rs1,422, Azad trades at ~106x FY25 EPS — elevated but compressing rapidly to 68x FY26E and 39x FY28E as earnings scale. Howmet has re-rated from 23x (FY20) to 65x (currently) as it has consistently delivered double-digit EPS growth. The same dynamic should play out for Azad as it demonstrates delivery across 3–4 more quarters.

P/BV: Azad's ~6x P/BV is now directly comparable to Howmet's ~10x P/BV — validating that the market is already treating Azad as a high-quality precision manufacturer deserving comparable book-value multiples.

P/FCF: Azad is in a pre-FCF stage — a direct consequence of aggressive capacity investment. Howmet itself was FCF-negative during peak capex (2017–2019). Howmet now generates US$1.4bn+ FCF annually at 93% net income conversion. Azad's FCF inflection (turn positive) is expected in FY27–FY28, at which point P/CF multiples will be directly comparable.

PEG check: At 60x PE on ~34% EPS CAGR (FY25–FY28), Azad's PEG = 1.76x. Howmet's current PEG = ~2.0x. Investors are paying less per unit of growth for Azad than for its global peer — the higher nominal PE is more than offset by faster earnings growth.

4. Comprehensive Peer Scorecard

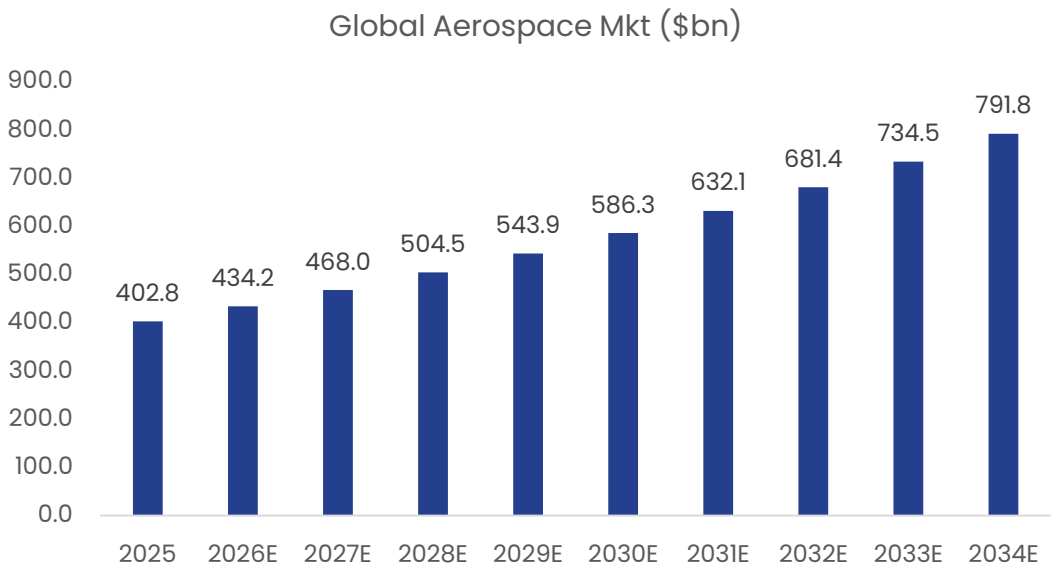
Parameter	Azad Engineering	Howmet Aerospace	Assessment
Revenue (FY25/FY25)	Rs457 Cr (~$41M)	$7.43 Bn	Howmet is ~180x larger. Azad is early-stage; scale gap narrows over 5–7 years.
Revenue Growth (4Y CAGR)	~39% (FY21–25)	~13% (FY21–25)	Azad growing 2.7x faster — early-stage scaling advantage driven by new plant ramp-ups.
Revenue Outlook	25–35% guided FY25–28	8–10% expected	Azad's growth runway significantly longer; driven by capacity additions.
EBITDA Margin (FY25)	35.3% → 36%+ guided	25.8% (FY25)	Azad's margins already exceed Howmet due to India's structural cost advantage.
EBITDA CAGR (4Y)	~54% (FY21–25)	~18% (FY21–25)	Azad's EBITDA leverage reflects operating scale-up from a low base.
Balance Sheet Size	Rs797 Cr FY24 (Rs1,855 in FY25)	$10.7 Bn FY24	Howmet ~113x larger. Azad growing rapidly (4.6x total assets in 4 years).
Balance Sheet Quality	Net cash post-IPO; low debt	ND/EBITDA ~1.4x; deleveraging	Both healthy. Azad well-capitalised for next growth phase.
ROCE	10–12% (FY23–24); dip FY25	ROIC ~14%; ROE ~30%	Howmet higher; Azad's dip is temporary (capex dilution). Recovery expected FY26–27.
P/E Multiple (FY25)	~106x FY25	~54x trailing	Azad premium reflects faster EPS growth (~50% CAGR). PE compresses to 38x by FY28E.
P/BV Multiple (FY25)	~6x	~10x (current)	Comparable P/BV — validates similar valuation treatment at scale.
P/FCF	Negative (growth capex)	~52x (record FCF)	Azad pre-FCF stage; FCF inflection expected FY27–28 as capex normalises.
PEG Ratio (fwd)	~1.8x (60x PE / 34% growth)	~2.0x (current)	Azad is cheaper per unit of growth than Howmet despite higher nominal PE.
Sector Tailwinds	Aero engines + IGT + O&G	Identical	Both benefit from global aero engine upcycle; data-centre IGT demand accelerating.
Key Risk	New plant ramp-up execution	Boeing/Airbus delays	Azad risk is internal/controllable; Howmet risk is OEM-dependent.

Valuation conclusion: At CMP of Rs1,422, Azad trades at 39x FY28E earnings — in line with early-stage Howmet multiples when it was at a comparable point in its growth trajectory. The initiation target of Rs2,193 (60x FY28E) represents a natural re-rating point that: (a) is already exceeded by Howmet's current 65x multiple, (b) implies a PEG of only 1.8x vs Howmet's 2.0x, and (c) is supported by 25–35% revenue growth visibility backed by contracted order flows. The Howmet precedent makes 60x FY28E not just justifiable — it is, in fact, conservative if Azad executes on its capacity ramp-up.

Industry Analysis:

Global Aerospace industry overview:

The Aerospace sector was valued at US$402.75bn in 2025 and is expected to reach US$791.78bn by 2034. Out of this, the Commercial Aerospace sector is valued at US$230bn in 2025 and is projected to clock CAGR of 5% over the next decade. This growth is driven by the resurgence in aviation demand, which has occurred post COVID-19. As the world returns to normalcy, demand for air travel recovered to pre-COVID levels in CY25 and is expected to grow further owing to various factors such as increasing disposable incomes, global connectivity and recovery of business. Air Cargo demand, though volatile, is expected to grow and reach 101.13mn tonnes by CY30. With steady demand for aviation, airlines globally have ordered many aircrafts. The total fleet size stood at 27,000 aircrafts in 2022, which is expected to reach 48,575 by 2042. Most of the new orders are being led by narrow-body aircrafts as more budget carriers enter the fray.



As the number of aircraft ordered reaches new heights, so does the order backlog. In 2024, worldwide aircraft backlog stood at 17,000, higher than the average backlog of 13,000 aircraft per year from the previous decade. Aircraft deliveries have not been able to make a recovery; in 2024, the total aircraft delivered were 1,226 compared to 1,374 in 2019. This gap in deliveries points to supply chain disruption, which has been caused by various factors. Post COVID-19, most workers in the western world did not return to work, which led to a lack of skilled manpower. Apart from that, global trade and geopolitical concerns have also disrupted supply chains. To mitigate this issue, various OEMs have started outsourcing key operations and diversifying their value chain globally.

Evolution of the Aerospace industry:

1960s and 1970s:
The Vertically Integrated Era

- Airframers around the world have vertically integrated operations
- Supply chains short and local for Airframers
- Indian aerospace dominated by HAL, focusing largely on defensive aircraft

1980s and 1990s:
Post Cold-War

- Trade liberalization reforms around the world
- Airframers look to expand operations globally
- Tier-1 suppliers take on engineering and risk-sharing roles
- India sees growth in IT and Engineering talent; starts laying foundation for future of the Aerospace industry

2000s:
Supply Chain Consolidation

- Airframers divest aerostructure manufacturing units, enabling the rise of Tier-1 suppliers
- Risk-sharing becomes the industry norm, with suppliers entering MRO to recoup R&D investments
- Complex aircraft programs deepen supply chain integration

2004-2006:
Indian Aerospace sees an inflection point

- With a declining Auto and IT industry, the laid off workforce gets absorbed by the emerging Aerospace industry
- Launch of airlines like IndiGo and Kingfisher besides new orders make India a promising aviation market
- OEMs see the first wave of global outsourcing

2010s:
Expansion, Digitalization and Rising Traffic

- With air traffic doubling almost every 15 yrs, airframers see skyrocketing orders
- A320 & B737 families see a surge in demand
- A value-shift seen in avionics, electronics & advanced materials
- Rise of MRO as a key business model, with more OEMs looking to recoup their investments

2020-2021:
The COVID-19 shock

- Global aviation demand collapses, with ~60% contraction in air travel
- Thousands of skilled workers exit the industry, creating a labor shortage for future
- Supply chains see disruption, with global shortages for RMs and components

2022-2023:
A Resurgence

- Air travel demand sees a shocking recovery: demand forecast in 2026-2027 materializes in 2022-2023
- OEMs face parts shortages and workforce exit, delaying production
- Aircraft backlogs soar to as high as 10-12 years
- India sees an inflection point: with record number of airplanes being ordered by domestic carriers
- OEMs look to move out of Europe, with a Europe +1 strategy, making India a key source in future

2024-2025:
Structural transformation and digitalization

- With historic backlogs, demand for MRO surges as airlines look to increase aircraft lifespans
- Forecasting and production planning still unreliable, and increasing TATs cause large delays
- The role of Tier-2 and Tier-3 suppliers increases, as the industry sees a structural change, a trend like the one seen in Tier 1 Engine OEMs earlier
- New MRO hubs set up in India
- Indian aerospace exports touch US$6.9bn in 2025

Understanding the supply chain:

Basic Value Chain:

Airframers

AIRBUS

BOEING

OEM's

ROLLS ROYCE

PRATT & WHITNEY DEPENDABLE ENGINES

SAFRAN AEROSPACE · DEFENCE · SECURITY

Component Manufacturers

AZAD

PTC INDUSTRIES ASPIRE · INNOVATE · ACHIEVE

DYNAMATIC TECHNOLOGIES

Stage 1: Raw Material Procurement (Tier 4)

This stage involves sourcing high-performance materials such as titanium, nickel alloys, aluminum and composites. These materials are produced by specialized suppliers like ATI, Doncaster's and CPP, who deliver aerospace-grade metals and castings that meet stringent quality standards.

Stage 2: Component & Detailed Part Manufacturing (Tier 3)

Tier 3 suppliers convert raw materials into precision components through machining, forging, molding and composite fabrication. Companies such as Cadence Aerospace and Greene-Tweed manufacture detailed parts that serve as the building blocks for larger assemblies.

Stage 3: Subsystem Assembly (Tier 2)

Tier 2 suppliers integrate multiple components into functional subsystems—landing gear modules, engine casings, wing substructures, hydraulic assemblies, etc. Key players include Magellan Aerospace and Barnes Group, who supply certified sub-assemblies to Tier 1 integrators and OEMs.

Stage 4: Aerostructure & System Integration (Tier 1)

Tier 1 suppliers assemble major aerostructures such as fuselage sections, wings, nacelles, and propulsion systems. Engine OEMs like Rolls-Royce and GE Aerospace, along with system suppliers such as Honeywell and Collins Aerospace, provide critical integrated systems ready for final assembly.

Stage 5: Final Aircraft Assembly

Airframers such as Boeing and Airbus integrate all major structures, engines, avionics and interiors into the final aircraft. This includes full-system testing, certification processes and delivery to airline customers.

Stage 6: Aftermarket MRO

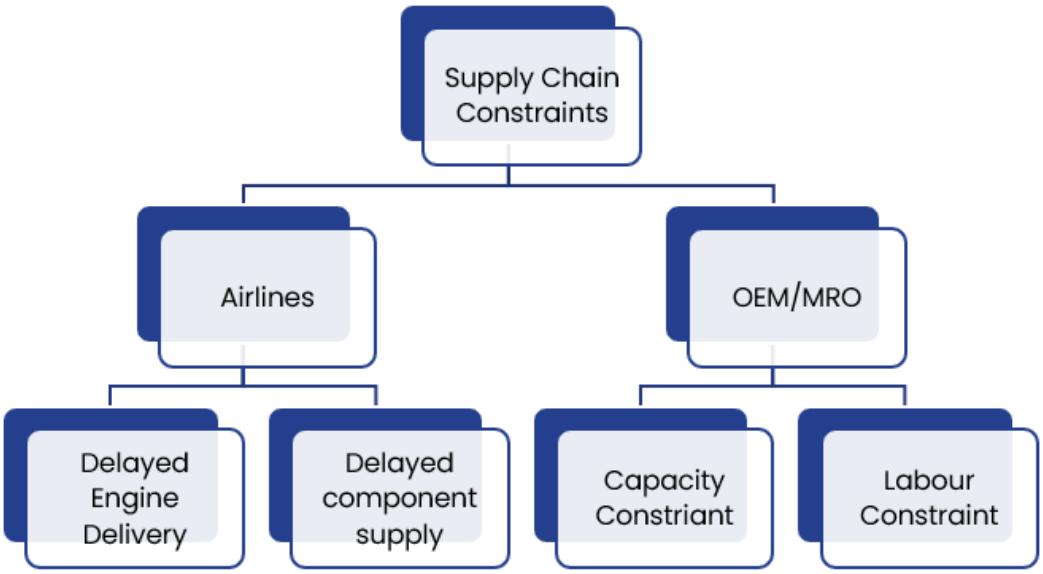
Post-delivery, aircrafts undergo continuous maintenance and overhaul. The Aftermarket segment is critical for ensuring airworthiness and generates substantial long-term revenue for OEMs and service providers.

What's happening in the Aerospace industry?

The highly consolidated, tiered structure of the commercial aircraft industry has found it difficult to absorb multiple recent and overlapping market shocks — from the disarray caused by COVID-19 to geopolitical conflicts-driven material shortages and tight labour markets. As a result, airlines are waiting for new aircrafts with lower fuel consumption while facing higher maintenance & repair costs for an aging fleet.

1. **Supply chain constraints:**

   The Aviation sector cannot function without a reliable supply chain. Yet today, grounded aircrafts, delayed deliveries and escalating maintenance & leasing costs are clear symptoms of a system under strain.



In 2024, commercial **aircraft backlogs** hit historic highs, with over **17,000 aircrafts** on order, equivalent to more than 12 years of production at current rates. The backlog and time to work through it have increased significantly since early 2000s and have further increased post-pandemic. Despite strong demand, major airframers remain far below their peak annual deliveries: Airbus delivered 766 aircrafts in 2024 (down from a 2019 high of 863) while Boeing's deliveries dropped sharply, from 806 in 2018 to 348 aircrafts in 2024.

The existing backlog, coupled with lower production, has resulted in longer overall lead times. Aircraft delivered in 2024 took 6.8 years to make it to the airlines, up from 4.5 years in 2018. For airlines, this means delayed fleet renewal plans and constrained capacity growth. This has led to older aircraft remaining in service longer than planned, delaying their retirements and increasing operational costs.

2. **Unstable aviation economic model:**

   With engine and equipment system, OEMs are receiving a growing share of profitability from **aftermarket repairs and spare parts**, rather than new equipment sales. Timely access to serviceable parts and maintenance is essential to keep aircraft flying and avoid unnecessary groundings. This requires restoring standard supply lead times and repair turnaround times. A stronger supply chain also depends on transparency and

collaboration across the value chain. Without decisive action, these bottlenecks risk constraining aviation growth and sustainability.

3. **A generational transition affecting aviation industry:**

The Aerospace industry's supply chain delays, quality challenges and production shortfalls are increasingly rooted in a structural shortage of skilled labour rather than weak demand. Even before COVID, an aging workforce and limited inflow of younger, technically trained workers had strained the system. The pandemic accelerated the problem, as large-scale layoffs in 2020 led many experienced technicians and mechanics to retire early or exit the industry, permanently eroding institutional knowledge. Today, the workforce is younger but less experienced, raising execution and quality risks in a precision-critical industry. This has resulted in persistent shortages of highly skilled roles across the US, Europe and parts of Asia—despite significant wage increases—forcing aerospace companies to lean more heavily on automation, process discipline and capable Tier-1 suppliers to sustain production.

4. **Increasing length of time for aircraft certification**

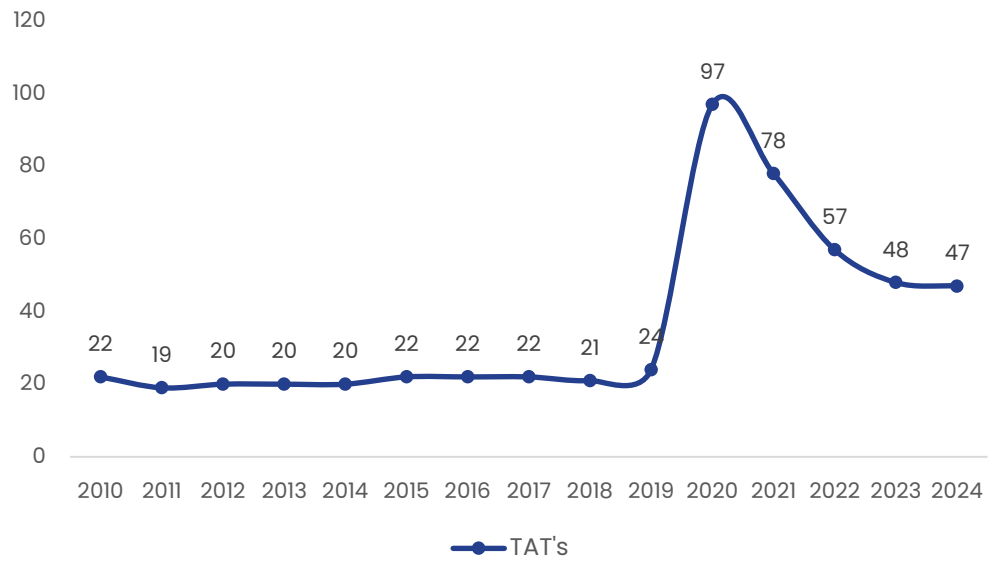
Historically, certification times ranged ~1-2 years. New-generation platforms have tended to be at the higher end of this range, but noticeably, Boeing certifications of the 737 MAX 7/10 and 777x are now in their fourth and fifth years, respectively, and it could take until 2026 for these platforms to be certified. These platforms account for 14% and 24% of the narrowbody and widebody orderbooks, respectively. Several issues collectively may be impacting certification timelines, including but not limited to OEM design changes, airline interior customization and regulator capacity. Many regulators, including the Federal Aviation Administration (FAA) and European Aviation Safety Agency (EASA), are facing staffing shortages and challenges in attracting specialised talent. For example, in 2023, a reported 40% of FAA certification engineers had less than two years of experience. FAA processes also frequently require paper documentation, as opposed to digital copies. In addition, evolving regulations require more comprehensive documentation and substantiation, compelling manufacturers to revisit previous work.

5. **Increased maintenance turnaround times**

Following the surge in aircraft backlogs and extended delivery times, the Aviation industry is facing widespread increases in TATs, further constraining airline operations. For engines in particular, turnaround times have seen sharp increases from the industry standard of **30-60 days**. Shop visits are lasting 300+ days in some instances, with full overhauls **taking around 75 days** and quick turns taking around 50 days. Similarly, landing gear overhauls typically lasting 45 days under normal conditions are now taking 90-120 days, and this trend can be seen across numerous other components, including APUs. Piece part availability was the top issue — outpacing supply/demand imbalances, insufficient labour and OEM control of key repairs.

Longer TATs lead to more inventory in the repair cycle compared to the shelf and available for use by airlines. This means airlines either need to hold more inventory to have the same amount on hand as they had previously, or they need to accept a deteriorating performance. And with less inventory on the shelf, parts shortages become an issue, increasing the duration of aircraft on the ground, which requires parts to

resolve and increasing costs if alternative parts must be sourced at the last minute. Increased TATs and insufficient spares are ultimately reducing operational capacity, resulting in higher number of aircraft in storage (defined by aircraft that are 'out of service' for at least seven consecutive days). While the pandemic led to a spike in stored aircraft in 2021 and 2022, the overall figure remains significantly higher than historical norms even as air travel demand has returned.



But, out of all, the only Aerospace segment which is benefitting is MRO

MRO includes all activities that are required to keep an aircraft running, safe and worthy of flying. It covers activities ranging from routine checks to overhauling major components like engines and landing gear. OEMs can be classified broadly into four categories.

1. **Line maintenance:** Line maintenance refers to routine, day-to-day aircraft checks carried out at the airport between flights to ensure the aircraft is safe and ready for operation. It includes tasks like visual inspections, fluid top-ups, basic troubleshooting and minor component replacements. Since these activities are performed with minimal downtime, line maintenance is essential for keeping aircraft turnaround times short and maintaining high fleet availability.
2. **Engine MRO:** Engine MRO is concerned with the servicing, fixing and overhaul of aircraft engines. It is mainly concentrated in the hands of OEMs whereas in other MROs, the market is more fragmented. In Engine MRO, the role of OEMs changes in line with the lifecycle of the engine platform. In the development and Entry Into Service (EIS) phase, OEMs typically have complete control over the MRO. As the lifecycle enters the growth phase, OEMs start opening doors to joint ventures and licensing to other MRO firms and expanding capacity. In the second last phase, MRO work can be accessed through a broad range of MRO firms, and finally in the sunset phase, the market focuses on harvesting and repair and the usage of Used Serviceable Materials (USM) and PMA become good alternatives to new OEM parts.
3. **Component MRO**: Component MRO is concerned with critical aircraft parts such as landing gear and avionics. The component MRO market is more fragmented compared to Engine MRO, with OEMs having the most impact in high value items, such as avionics

and landing gear. MRO for Buyer Furnished Equipment (BFE) items is negotiated bilaterally between airlines and OEMs, whereas for Seller Furnished Equipment (SFE) items, terms of product support are comprehensive and available to OEM vendors by the airframers.

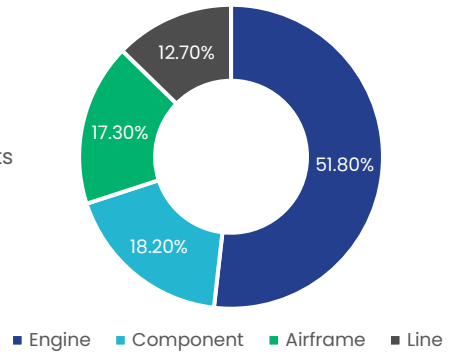
4. **Airframe MRO**: Airframe MRO is concerned with specialized services for an aircraft's structures such as fuselage and wings. The market for airframe MRO is fragmented, with airlines themselves and independent providers dominating the market. Firms range from single hangar providers to global providers available across regions. OEMs have a very low share in airframe MRO as they don't have a competitive advantage to enter this business. The only exception is regional jets, where Embraer and MHIRJ hold 70%+ market share.

By the end of the current forecast period, total global MRO spend will reach US$156bn, up more than 31% from 2025 and 46% above its previous peak in 2019. As a result of the super cycle, the industry will see long queues and increased turnaround times at MRO facilities, where capacity across the board was already tight, particularly for engine MRO.

A significant driver of rising demand in the Aerospace industry is the sharp increase in maintenance requirements stemming from an aging and heavily utilized global aircraft fleet. The average age of aircraft in service climbed to 13.4 years in 2024 from 12.5 years in 2023, marking an unusually steep one-year increase and leading to higher servicing & parts replacement to maintain safety and reliability. At the same time, undersupply of new aircraft has pushed existing fleets to fly harder, with average annual flight hours per aircraft rising ~15% to ~2,800 hours in 2024 from ~2,400 hours in 2022. Higher utilisation directly translates into greater maintenance intensity, particularly for engines, components and line maintenance. As a result, global aircraft utilistion surpassed 78mn flight hours in 2024, exceeding the previous 2019 peak. This is expected to grow at ~3.4% annually, outpacing fleet growth and driving sustained MRO demand over the next decade.

The dual effect of older fleets remaining in service longer and newer fleets requiring maintenance sooner than anticipated (to address reliability and durability issues) have created a "super cycle" for the Aftermarket segment.

MRO Expenditure by Segments



More than 90% of MRO services are currently outsourced to Sri Lanka, Singapore and the UAE. India, while self-reliant in line maintenance activities, is still dependent on other countries for most of its MRO needs. With MRO outsourcing increasing by the day, there is an opportunity for India to become a global MRO hub in the region. MRO hubs are being set up by various companies, such as by Safran in Hyderabad, Boeing in Nagpur and Air India-Pratt and Whitney JV in Mumbai.

Impact on overall Aerospace industry:

The Aviation industry has rebounded strongly from the pandemic shock, with global passenger traffic expected to reach ~9.9bn in 2025 (~104% of 2019 levels; IATA). Structural growth in air travel continues to support fleet expansion, with the global commercial fleet projected to rise to ~38,300 aircraft by 2035. Over the longer term, Boeing estimates the fleet will approach ~50,000 aircraft by 2044, driven by both traffic growth and replacement demand (~21,000 aircraft over 2024–2044). Demand remains heavily skewed toward narrowbody platforms, with single-aisle aircraft expected to account for ~75–76% of total deliveries.

Backlog-driven cost pressures

Despite strong demand, supply-chain constraints and OEM production delays continue to create operational inefficiencies across the Aviation industry:

1. **Fuel inefficiency:**
   Ongoing OEM delivery delays are forcing airlines to retain older-generation aircraft longer than planned. These aircraft have materially higher fuel burn versus new gen narrowbodies (neo/MAX), resulting in incremental fuel costs of ~US$4.2bn as carriers operate less efficient fleets while waiting for backlog normalization.

2. **Maintenance costs:**
   The aging global fleet and elevated utilization levels are increasing maintenance intensity, with more frequent heavy checks, higher component replacement cycles, and elevated shop visits. This is translating into incremental maintenance costs of ~US$3.1bn across airlines.

3. **Engine leasing:**
   Supply-chain disruptions and durability issues (particularly across certain new-generation engine platforms) have extended engine turnaround times, compelling airlines to lease spare engines to maintain operational continuity. This has driven excess leasing costs of ~US$2.6bn in 2025, alongside a ~20–30% increase in aircraft lease rates vs. 2019 levels.

4. **Inventory costs:**
   Persistent parts shortages and supply volatility have prompted airlines to increase buffer inventory of critical spares and rotables to avoid operational disruptions. This precautionary stocking is adding ~US$1.4bn in incremental inventory holding costs.

Indian Aerospace Value Chain

Impact on the Indian Aerospace industry:

India is among the world's top three fastest-growing aviation markets. Domestic passenger volume reached 161.3mn in CY24 and IATA projects annual traffic will triple to 425mn by 2044. With only ~4% share of global air traffic compared to ~17% for China and ~18% for the US, India remains structurally underpenetrated.

This growth is underpinned by:

- Low-cost carriers are driving affordability
- Rising disposable incomes and an expanding middle class
- UDAN scheme, which has created a new category of regional flyers through subsidised connectivity
- Fleet expansion: India's fleet has more than doubled in the past decade to ~890 aircraft, with 2,325 aircraft on order, one of the largest order books globally
- Airport expansion: 50 new regional airports are planned in the next five years (current: 147).

These factors together create a **sustained demand environment for regional and narrow-body aircraft**, directly translating into higher demand for high-precision components.

Why are global OEMs coming to India?

1. **Unprecedented aircraft backlog**
   Global aircraft OEMs are facing an unprecedented backlog of **~14 years** of aircraft orders, driven by strong passenger demand and limited production capacity. To accelerate deliveries and de-risk execution, OEMs are expanding manufacturing footprints beyond traditional geographies.
2. **Supply chain re-orientation**
   Post-COVID disruptions and geopolitical risks have pushed OEMs to diversify and regionalise supply chains, reducing dependence on single-country sourcing. India offers a stable, scalable and increasingly qualified alternative within global aerospace supply networks.
3. **Structural cost advantage**
   India provides a significant cost advantage, with **labour costs ~70% lower than the US and 30–40% lower than China**, while maintaining improving quality standards. This allows OEMs to structurally lower production costs without compromising on precision or reliability.
4. **Deep STEM talent pool**
   India has a large and growing base of STEM-trained engineers and technicians, supported by excellent engineering institutes and vocational programs. This talent availability supports long-term scaling, complex manufacturing and knowledge-intensive aerospace work.

The Week

GE Aerospace launches Next Engineers program in Bengaluru to power India's STEM future

It is a four-year college and career readiness program that will reach 4,000 students to expand the engineering talent pipeline. The initiative...

The Hindu

4,000 students in Bengaluru can join GE global STEM training programme

GE Aerospace Foundation has expanded its global career-readiness programme, Next Engineers, that is aimed at building a pipeline of STEM...

The New Indian Express

Aviation, Aerospace and Defence EduCity to come up at Andhra's Bhogapuram

Spread over about 136 acres, the EduCity will offer education and training in aviation, aerospace and defence sectors and seek to position...

Times of India

In a first, Adani, Embraer tie up to build commercial fixed-wing planes in India

India Business News: NEW DELHI: India will soon join a select group of nations that have a final assembly line (FAL) for commercial...

Aerospace Pioneer Natilus Expands Global Reach to Meet Demand from One of Today's Fastest-Growing Aviation Markets, India

PRNewswire/ -- Natilus, a U.S. aerospace manufacturer of blended-wing body aircraft, today debuted Natilus India, a subsidiary headquartered...

The Energy industry:

The Energy Equipment sector is **entering a structural, multi-decade upcycle**, underpinned by electricity demand, grid stress and baseload reliability needs. The market is highly concentrated, with GE Vernova, Siemens Energy and Mitsubishi Power controlling nearly two-thirds of global construction. GE Vernova alone has 55GW of turbines under construction. Asia drives growth: Two-thirds of gas-fired capacity under construction is in Asia, led by China (151 GW in development and 46 GW under construction). Nearly 47% of turbines under construction (82 GW) are hydrogen-ready, reflecting a shift toward cleaner fuels. Global OEMs' commentary reinforces that this cycle is larger, longer and more profitable than prior cycles.

For component manufactures, with exposure to the Energy segment, this translates into:

- High order visibility
- Stickier OEM relationships
- Rising share of wallet per platform
- Long runway for margin-accretive growth

Why is the Energy Equipment & Power Infrastructure cycle entering a multi-year growth phase?

1. **Structural surge in electricity demand (largest in decades):**
   The global power demand growth is no longer cyclical but structural, driven by:
   - AI data centres & hyperscalers with high, volatile base-load requirements
   - Re-industrialisation in the US and select global markets
   - Electrification of transport, buildings and industry

   Electricity demand is rising at a scale not seen in decades, setting up a longer and larger capex cycle rather than a short-term spike.

2. **Gas Power as the "Bridge Baseload" of the energy transition**
   Despite high renewable energy penetration, gas turbines are indispensable due to:
   - Grid instability at high RE penetration
   - Need for fast-ramping, reliable baseload for data centres
   - Delays in transmission, permitting and fuel availability

   For instance, GE has ~80GW of gas turbines already on contract and it is ramping up production to 20GW annualised by FY26; it expects to be sold out for 2030 deliveries by end-2026

3. **Electrification & Grid capex: A parallel growth engine**
   Electrification backlog is expected to double again by 2028, driven by:
   - HVDC transmission for RE integration
   - Grid resilience & reliability projects
   - Hyperscalers-driven direct procurement of grid equipment

   This segment is now the fastest-growing business. OEM capex confidence across power systems strengthens long-term supplier capex, localisation and outsourcing intensity, indirectly supporting wallet-share expansion.

4. **Optionality from SMRs, Nuclear & Next-Gen technologies**

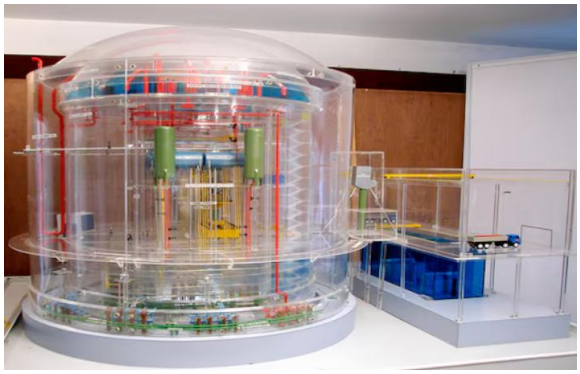
   While not in near-term numbers, the global OEMs are investing aggressively in:

   - Small Modular Reactors (SMRs)
   - Carbon capture, fuel cells and solid-state transformers

   These programs add embedded long-term optionality to the energy ecosystem without requiring incremental downside risk today.

Nuclear Power Market:

Global nuclear power generation is set to grow by ~2% annually during 2025–2026, supported by reactor restarts in Japan, new builds in China and India, and steady output from the US and France. Nuclear power's role in energy diversification and de-carbonisation is expanding. India targets 22,000MW of nuclear power capacity by 2050, with multiple new reactors under construction and collaborations with Russia, France and the US. Indigenous programs such as the Advanced Heavy Water Reactor (AHWR) and future thorium-based designs underscore India's long-term ambitions.



AHWR

Steam / Thermal Turbine Market:

The global steam turbine market was valued at US$17–19bn in 2024 and is expected to grow modestly at ~2.4–2.5% CAGR to reach US$21–23bn by 2030–2034. In India, growth is stronger, with the market valued at US$934mn in 2024 and forecast to reach US$1.4bn by 2033 (CAGR ~4.3%). Another estimate places it at US$3.3bn in 2022, rising to US$4.4bn by 2030 (CAGR ~3.5%). Separately, the steam turbine service market is valued at US$19.5bn in 2024 and is expected to reach US$31.8bn by 2034 (CAGR 5%) — larger than the turbine equipment market itself, reflecting the long-term value in MRO services.

Oil and Gas Industry:

The global Oil and Gas industry is witnessing a **cyclical recovery on the back of strong energy demand, geopolitical energy security concerns and renewed investments in hydrocarbon infrastructure**. While long-term energy transition policies continue to promote renewable energy, oil and natural gas remain critical to the global energy mix, particularly for transportation, petrochemicals and baseload power generation. As a result, the sector is expected to see a sustained capex across upstream, midstream and energy infrastructure segments over the coming decade.

One of the key tailwinds for the industry is the continued growth in global energy demand, particularly from Emerging Markets such as India, Southeast Asia and parts of Africa. Rising industrialization, urbanization and mobility needs in these regions are expected to drive incremental demand for hydrocarbons. Global oil demand has already surpassed pre-pandemic levels and continues to grow steadily while natural gas demand is expanding due to its relatively lower carbon intensity compared to coal and oil.

Another important driver is the growing role of **natural gas as a transition fuel in the global energy mix.** Many countries are increasingly shifting from coal-based power generation to gas-fired power plants to reduce emissions while maintaining energy reliability. This transition is **driving investments in gas infrastructure, including LNG liquefaction plants, regasification terminals, pipelines and gas turbine-based power generation systems**. The expansion of the global LNG trade, particularly in Asia and Europe, has significantly increased demand for gas turbines, compressors and related rotating equipment.

Energy security concerns following **geopolitical disruptions have also prompted many countries to diversify their energy supply chains and increase domestic hydrocarbon production**. Governments and energy companies are investing in upstream exploration, offshore drilling projects and production capacity expansion to reduce reliance on imports. This has resulted in higher spending on oilfield equipment, industrial turbines and high-performance components used across drilling and production systems.

In addition, the **modernization and efficiency improvement of existing oil and gas infrastructure** is creating incremental demand for advanced engineering components. Industrial gas turbines used in energy generation, and oil and gas processing facilities are undergoing continuous technological upgrades aimed at improving efficiency, durability, and emissions performance. These upgrades require high-precision components made from specialized alloys capable of operating in high-temperature and high-pressure environments.

From a supply chain perspective, the Oil & Gas industry is characterized by high entry barriers due to stringent qualification requirements, long certification cycles and the need for advanced manufacturing capabilities. Suppliers of critical components such as turbine blades, airfoils and compressor parts must meet strict quality standards set by global OEMs, often requiring multi-year qualification processes. As a result, the global supplier base for such components remains highly concentrated.

Overall, sustained hydrocarbon demand, expansion in global LNG infrastructure, increasing adoption of gas-based power generation and high "barriers to entry" in the component supply chain collectively act as structural tailwinds for the Oil & Gas equipment manufacturing ecosystem. These factors support a favorable long-term growth outlook for specialized engineering companies supplying critical components to the global Oil & Gas industry.

Outlook and Valuation: Initiate with BUY

Backed by strong revenue growth visibility, Azad Engineering is well positioned to sustain its growth momentum, supported by its robust order book and long-term contracts with global OEMs. The company operates in a niche segment of manufacturing precision, mission-critical and zero-defect components, where lengthy qualification cycles create high entry barriers and ensure strong customer stickiness and multi-year revenue visibility. Additionally, commissioning of dedicated lean facilities, ongoing capex plans aligned with confirmed demand, and structural cost advantages of ~30% over global peers further strengthen its competitive positioning and scalability.

At CMP of Rs1,471, Azad Engineering trades at ~40x FY28E EPS. We initiate coverage with a BUY and Target Price of Rs2,193, valuing it at ~60x FY28E EPS. We believe this valuation is justified given Azad's strong execution track record, high-margin precision manufacturing capabilities, deep integration with global OEM supply chains and long runway for growth, driven by Aerospace, Defence and Energy sectors.

Key risks:

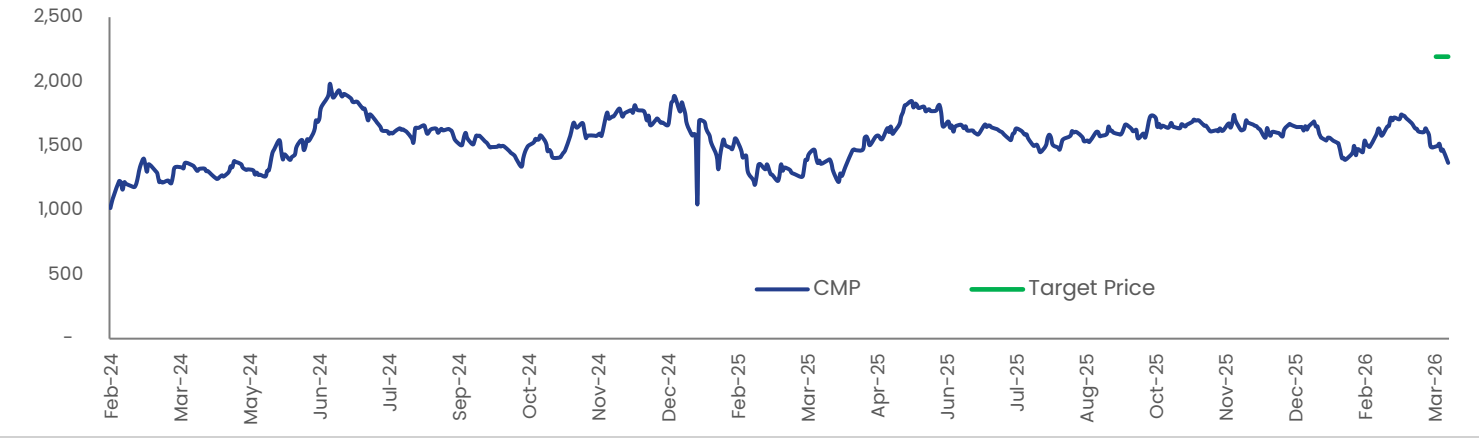
- Promoter/key person risk
- Disqualification risk (loosing OEM approval)
- Execution risk
- Energy transition risk

Recommendation History

Date	CMP (Rs)	TP (Rs)	Rating	Analyst	Report Link
25-03-2026 (IC)	1,471	2,193	BUY	Koushik Mohan	-

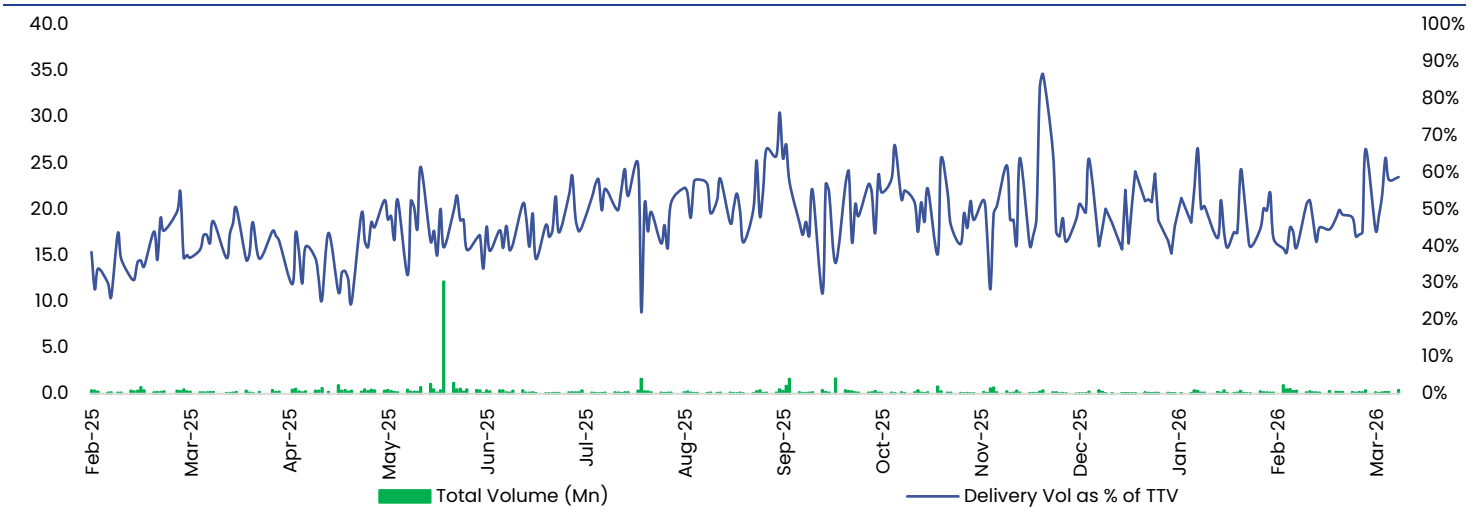
Source: Ashika Institutional Research

Historical market price and target price



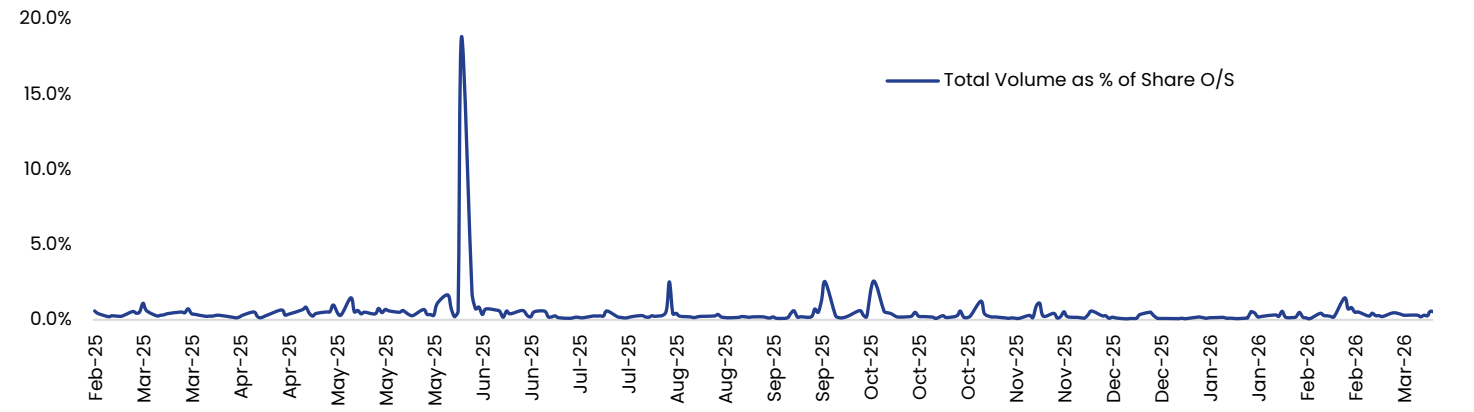
Source: ACE Equity, Ashika Institutional Research

Traded volume vs. delivery volume (1 Year)



Source: ACE Equity, Ashika Institutional Research

Traded volume as percentage of share O/S (1 year)



Source: ACE Equity, Ashika Institutional Research

ashika Institutional Equity Research

Financials & Valuation:

P&L Statement

Y/E Mar (Rs mn)	FY24	FY25	FY26E	FY27E	FY28E
Revenue	3,408	4,574	6,174	8,027	10,836
COGS	460	637	864	1,124	1,517
Employee Cost	743	929	1,235	1,525	1,950
Job work charges	194	364	494	602	867
Tools	219	188	216	241	325
Power and fuel	136	165	247	321	433
Stores and spares consumed	143	178	216	321	433
Other Expenses	348	500	679	883	1,192
EBITDA	**1,166**	**1,613**	**2,223**	**3,010**	**4,118**
EBITDA margin (%)	**34.2**	**35.3**	**36.0**	**37.5**	**38.0**
Depreciation	205	295	375	539	601
Interest expense	473	184	382	388	427
Other income	320	106	412	201	191
Profit before tax	**808**	**1,240**	**1,878**	**2,284**	**3,280**
Taxes	230	375	526	639	918
Net profit	**578**	**865**	**1,352**	**1,644**	**2,362**
Net Profit Margin (%)	**17.0**	**18.9**	**21.9**	**20.5**	**21.8**
EPS (Rs)	**9.8**	**13.4**	**20.9**	**25.5**	**36.6**

Balance Sheet

Y/E Mar (Rs mn)	FY24	FY25	FY26E	FY27E	FY28E
Equity capital	118	129	129	129	129
Reserves	6,332	13,799	15,151	16,795	19,157
Net worth	**6,451**	**13,928**	**15,280**	**16,924**	**19,286**
Provisions	40	52	52	52	52
Debt	372	2,436	2,292	2,511	2,771
Other non-current liabilities	20	345	345	345	345
Deferred tax liab (net)	211	274	274	274	274
Total Liabilities	**7,093**	**17,035**	**18,243**	**20,107**	**22,728**
Fixed assets	2,545	4,166	8,291	9,253	10,651
Capital Work In Progress	454	798	798	(2)	(2)
ROU Assets	30	194	194	194	194
Other Financial Assets	247	265	265	265	265
Other non-current assets	480	1,214	1,214	1,214	1,214
Net working capital	**3,337**	**10,399**	**7,481**	**9,184**	**10,407**
Inventories	1,330	1,893	2,778	3,652	4,876
Sundry debtors	1,700	2,235	3,334	4,334	5,851
Other current assets	599	872	872	872	872
Sundry creditors	500	801	1,187	1,557	2,124
Other current liabilities & Prov	380	770	770	770	770
Cash	589	6,970	2,454	2,653	1,701
Total Assets	**7,093**	**17,035**	**18,243**	**20,107**	**22,728**

Cash Flow Statement

Y/E Mar (Rs mn)	FY24	FY25	FY26E	FY27E	FY28E
Profit before tax	808	1,240	1,878	2,284	3,280
Depreciation	205	295	375	539	601
Tax paid	(146)	(175)	(526)	(639)	(918)
Working capital Δ	(993)	(817)	(1,599)	(1,504)	(2,174)
Other Operating adj.	56	(7)	382	187	236
Operating cash flow	**(69)**	**537**	**510**	**867**	**1,025**
Capital expenditure	(553)	(9,177)	(4,500)	(499)	(1,809)
Free cash flow	**(622)**	**(8,640)**	**(3,990)**	**368**	**(784)**
Equity Raised	2,400	7,000	-	-	-
Debt financing / disposal	(1,217)	1,920	(144)	219	260
Interest Paid	(473)	(154)	(382)	(388)	(427)
Net cash	**88**	**126**	**(4,516)**	**199**	**(952)**
Opening Cash/Bank Balance	**527**	**589**	**6,970**	**2,454**	**2,653**
Transfers From /To Bank	**(26)**	**6,255**	**-**	**-**	**-**
Closing Cash/Bank Balance	**589**	**6,970**	**2,454**	**2,653**	**1,701**

Ratio Analysis

Y/E Mar (Rs mn)	FY24	FY25	FY26E	FY27E	FY28E
Growth matrix (%)					
Revenue growth	35.4	34.2	35.0	30.0	35.0
Op profit growth	61.4	38.4	37.8	35.4	36.8
Profitability ratios (%)					
OPM	34.2	35.3	36.0	37.5	38.0
Net profit margin	17.0	18.9	21.9	20.5	21.8
RoCE	10.7	5.7	7.4	9.3	11.7
RoNW	13.6	8.5	9.3	10.2	13.0
RoA	20.7	21.1	18.7	17.6	23.3
Valuation ratios (x)					
P/E	151.6	110.6	70.8	58.2	40.5
EV/EBITDA	75.0	56.5	43.0	31.8	23.5
Pay-out (%)					
Tax pay-out	28.5	30.2	28.0	28.0	28.0
Liquidity ratios					
Debtor days	182	178	197	197	197
Inventory days	142	151	164	166	164
Creditor days	54	64	70	71	72
WC Days	271	265	291	292	290
Leverage ratios (x)					
Interest coverage	2.0	7.2	4.8	6.4	8.2
Net debt / equity (%)	(0.0)	(0.3)	(0.0)	(0.0)	0.1

Du-Pont Analysis

Y/E Mar (Rs mn)	FY24	FY25	FY26E	FY27E	FY28E
Tax burden (x)	0.7	0.7	0.7	0.7	0.7
Interest burden (x)	0.8	0.9	1.0	0.9	0.9
EBIT margin (%)	0.28	0.29	0.30	0.31	0.32
Asset turnover (x)	1.12	0.89	0.67	0.85	1.00
Financial leverage (x)	0.5	0.4	0.6	0.6	0.6
RoE (%)	**9.0**	**6.2**	**8.8**	**9.7**	**12.2**

Additional Description

Company Background	Established in 2008 by visionary entrepreneur, Mr. Rakesh Chopdar, Azad Engineering stands at the forefront of global precision manufacturing, specialising in highly engineered, complex, mission and life-critical components for the Aerospace & Defence, Energy, Oil & Gas and Industrial Technology sectors. The company is revolutionizing the global precision manufacturing industry by leveraging cutting-edge technology and state-of-the-art infrastructure. With over 17+ years of operational excellence, Azad serves as a trusted Tier 1 supplier to the world's leading OEMs across several countries and has established a comprehensive ecosystem designed to effectively meet their ever-evolving needs.
Presence	Hyderabad, India
Key Managerial Persons	▪ Mr. Rakesh Chopdar - Chairman & Managing Director ▪ Mr. Murali Krishna Bhupatiraju - Managing Director ▪ Ms. Jyoti Chopdar - Whole-time Director ▪ Mr. Vishnu Pramodkumar Malpani - Whole-time Director
Credit Rating (Aug'25)	CARE A (Long-term) / CARE A2+ (Short-term) by CARE Ratings
Auditors	V. Nagarjuna & Associates, Chartered Accountants

Source: Company, Ashika Institutional Research

Key Milestones

Year	Achievements
2015	Expanded machining capabilities and qualified new product lines; commenced supply of precision components to global energy OEMs.
2016	Strengthened customer relationships with key aerospace and energy clients; enhanced shop-floor automation and quality systems.
2017	Inaugurated exclusive manufacturing bay for Mitsubishi Hitachi Power Systems Industries, Ltd. to qualify new products, further increase capacity and focus on process innovation.
2018	Expanded portfolio of qualified parts across turbine airfoils and aerospace actuators; continued capacity ramp-up at Hyderabad facilities.
2019	Entered into a long-term contract with Honeywell International, GE Aviation and other key clients; commenced market and geographical expansion.
2020	Continued deepening of long-term contracts with Honeywell International, GE Aviation and other key clients amidst market and geographical expansions.
2021	Raised ₹600 million in first round to address up-coming growth; entered into a long-term contract with Honeywell International, GE Aviation and other key clients.
2022	Raised ₹2,173 million to give an exit to existing investor and to deploy growth-capex; onboarded marquee investors from the sports fraternity.
2023	Successful IPO listing on BSE & NSE in December 2023, raising Rs 740 Cr at Rs 524/share; crossed Rs 252 Cr revenues; portfolio grew to 1,400+ qualified parts; received CARE A- credit rating
2024	Inaugurated dedicated lean manufacturing facility for Mitsubishi Heavy Industries; signed USD 73.47 Mn contract with MHI for precision airfoils; aerospace revenues grew 84% YoY; portfolio expanded to 1,700+ qualified parts
2025	Inaugurated GE Vernova dedicated facility (7,600 sq. m.) in April 2025; won USD 16 Mn order from Honeywell Aerospace ISC USA; raised Rs 700 Cr via QIP; FY25 revenues: Rs 453 Cr with EBITDA margin of 36%

Source: Company, Ashika Institutional Research

Top 10 Shareholders: Non-Promoter (as per latest disclosure)

Name	Shareholding (%)
ICICI Prudential Asset Management Co Ltd/India	4.93
Emirate of Abu Dhabi United Arab Emirates	2.62
Nomura India Investment Fund Mother Fund	2.23
Vanguard Group Inc/The	1.81
Axis Asset Management Co Ltd/India	1.39
Nomura Holdings Inc	1.20
Ashoka WhiteOak ICAV	1.17
Carne Global Fund Managers Ireland Ltd	1.10
Kotak Mahindra Asset Management Co Ltd/India	1.10
WhiteOak Capital Asset Management Ltd	1.01

Source: Bloomberg, Ashika Institutional Research

About the promoter:

Mr Rakesh Chopdar's journey is a story of resilience shaped on the factory floor rather than in a classroom. Growing up in Hyderabad, he struggled academically and did not complete formal schooling beyond Class 10. At an age when many were still in school, he began working in his father's small fastener unit. Surrounded by machines, tools and metals, he developed practical engineering skills through observation, repetition and relentless curiosity. What he lacked in academic credentials, he compensated for with discipline and hands-on mastery.

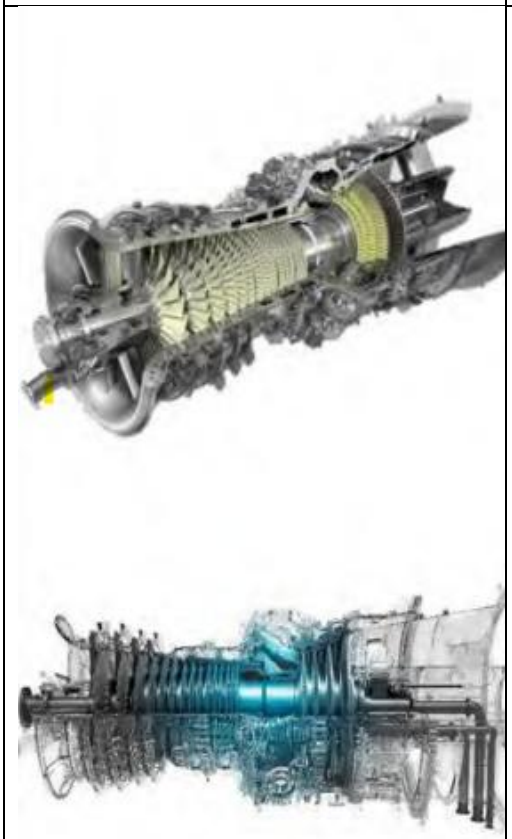
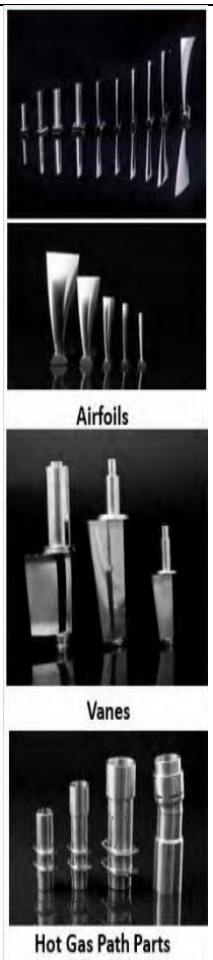
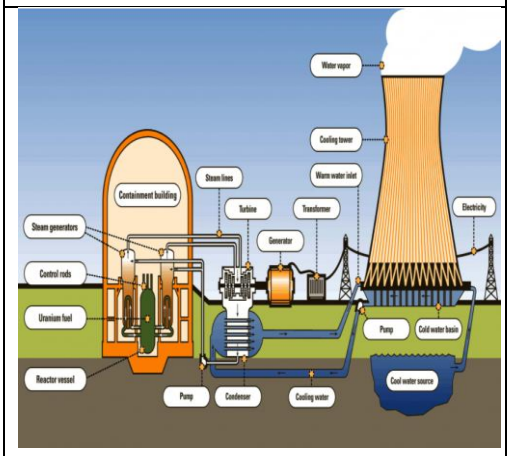
After more than a decade of learning the fundamentals of machining, quality control and customer relationships, Mr Chopdar set out on his own. In 2008, he founded Azad Engineering Limited in a modest industrial shed in Hyderabad with a single second-hand CNC machine. The company began by manufacturing precision turbine airfoils for thermal power applications — components that demand extreme accuracy and consistency. Winning the trust of global clients in such a high-precision field was not easy, but Mr Chopdar focused obsessively on quality, process discipline and long-term credibility.

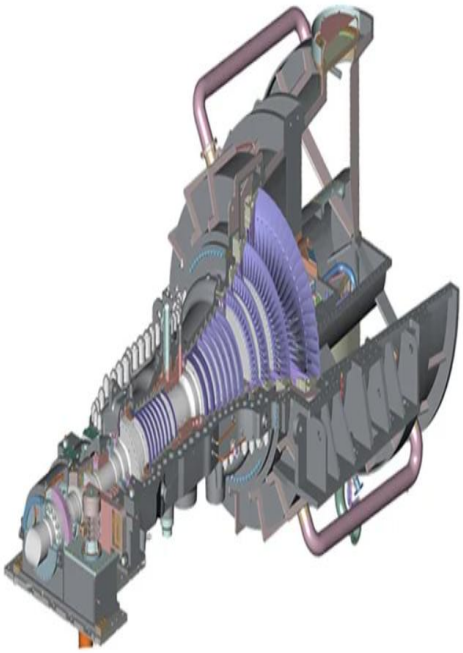
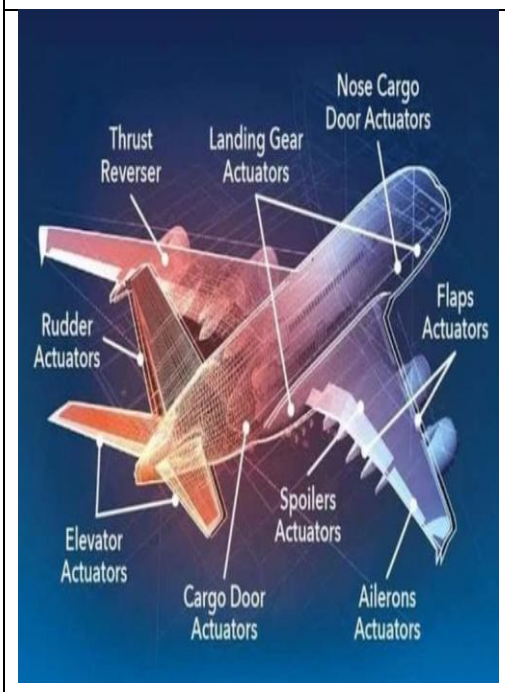
Over time, Azad expanded from Thermal Power into Aerospace, Defence and Advanced Energy Systems. The company went on to secure partnerships and supply relationships with global giants such as Rolls-Royce and Boeing, positioning itself as a critical supplier of complex rotating and structural components. What began as a small workshop evolved into a globally competitive precision engineering enterprise exporting to major international markets.

In 2023, Azad's growth story drew national attention when former Indian cricket captain Sachin Tendulkar invested in the company. Later that year, Azad went public, and its market valuation crossed the US$1bn mark — a remarkable milestone for a business founded by someone once written off for academic failure.

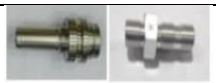
Today, Mr Chopdar is regarded as a self-made industrialist who built a global engineering platform from the grounds up. His journey reflects a belief that technical skill, persistence and uncompromising standards can transform early setbacks into an enduring success.

Product Portfolio

Application	Product Category	Our Products	End-Use Industry	Product Function
 Natural Gas Turbine	Turbine Airfoil Assembly	 Airfoils Vanes Hot Gas Path Parts	1. Electric Power Generation 2. Marine Propulsion 3. Industrial Applications Emergency Power Generation 4. Oil and Gas Industry 5. Renewable Energy Integration 6. District heating and cooling 7. Waste-to-energy	Gas turbines (Combustion turbines) are versatile machines that play a crucial role in various industries due to their efficiency, reliability, and flexibility. They operate on the principle of converting fuel energy into mechanical energy through combustion and then using this mechanical energy to generate electricity, propel vehicles, or drive other mechanical systems
 Nuclear Power Turbine	Turbine Airfoil Assembly	Airfoils Airfoils Nozzle Diaphragm LSB	1. Electricity generation 2. Power production 3. Baseload Power 4. Low environmental impact 5. Energy Security 6. Diverse Energy Portfolio 7. R & D	Nuclear turbine system is critical part of nuclear power plants, enabling the efficient conversion of nuclear energy into electricity. It's designed to operate safely, efficiently, and reliably while adhering to strict safety measures to prevent the release of radioactive materials into the environment.

			1. Electricity Power generation 2. Combined Heat and Power (CHP) plants 3. Geothermal Power Generation 4. Marine Propulsion 5. District Heating & cooling 6. Desalination Plants 7. Waste to energy Plants: Pulp and Paper Industry: Sugar Mills	Steam turbines are mechanical devices that convert thermal energy stored in steam into mechanical work, which can then be used to generate electricity, drive machinery, or perform various other tasks.
			1. Business Jet Aircraft 2. Commercial Aircraft 3. Defence Aircraft 4. Training Jet Aircrafts	Actuators in the aerospace market are used widely throughout aircraft, both fixed and rotary wing, as well as in missiles, spacecraft, and other marine and surface vehicles. Functions range from moving flight control surfaces to extending the footrests in business-class seats.

 	 Aero Engine Assembly	 Aero Airfoils Impeller, NGV Fan Blisk	1. Business Jet Aircraft 2. Commercial Aircraft 3. Defence Aircraft 4. Training Jet Aircraft	A jet engine turbine blade is individual component that makes up the turbine section of an aircraft engine Turbine Airfoils/blades function is to convert the energy of bypassing gas into the rotation of the rotor The force rotation in compressor Air foils/ blades of the rotor leads to air being sucked in and compressed by the Airfoils of multiple stages.
 APUs Aero Engines	**Air Generation and Valve Assembly in Aircraft Engine**		5. Business jet aircraft 6. Commercial aircrafts 7. Defence aircrafts 8. Training jet aircrafts	In aircraft engine valve and air generation systems facilitate the control of airflow in and around various functions of the aircraft, thereby regulating temperatures throughout the cabin and cockpit.

 Missile Systems	**Airframes and Booster**	 Aft End Skirt Fore End Skirt BB2KP Base B1 Igniter Body -1 B1 Igniter Body -2	9. Defence and missile manufacturing industries.	A missile is an intelligent unmanned rocket designed to carry the payload to a designated point with an aim of destroying the object/ target. The missile is designed keeping its target, trajectory, warhead, range, velocity and launch platform.
	Hydraulic System in Aircraft	 Nipples Adapters Tees Elbows End Fittings Caps	10. Business jet aircraft 11. Commercial aircraft 12. Defence aircrafts 13. Training jet aircrafts 14. Helicopters	Aviation hydraulic products are used in business jets, commercial jets, defence, training jet aircrafts and helicopters to help control and manage brakes, flaps, thrust reversers, flight controls, and landing gear. Hydraulic systems are the preferred system because they offer the perfect amount of pressure to operate these systems.

 Oil and Gas	**Up and Mid-Stream - Sub-Systems**		15. Oil and gas up-stream and mid-stream applications	Slips are a device used to grip and hold the upper part of a drill string to the drill floor on an oil rig. The slips are constructed as a collection of metal wedges, hinged together to form a circular shape around the drill pipe. On the inside surface, the slips normally have replaceable steel teeth that grips the pipe. In the oil and gas industry, a drill bit is a tool designed to produce a generally cylindrical hole (wellbore) in the earth's crust by the rotary drilling method for the discovery and extraction of hydrocarbons such as crude oil and natural gas.

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Website: www.ashikagroup.com

Recommendation & Absolute Return (w.e.f. Jan 01, 2025)

BUY	Expected positive return of 15% or above in 1-year horizon
ADD	Expected positive return of > 0% to 15% in 1-year horizon
REDUCE	Expected return of 0% to -15% in 1-year horizon
SELL	Expected to fall by >15% in 1-year horizon
UR	**UNDER REVIEW** - Rating and fair value, if any, have been suspended temporarily
CS	**COVERAGE SUSPENDED** - Ashika Institutional Research has suspended coverage of this company
NC	**NOT COVERED** - Ashika Institutional Research does not actively cover this company

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Juili Baviskar — Digitally signed by Juili Baviskar DN: cn=Juili Baviskar, o=Ashika Stock Services Ltd., ou=Institutional Equity Research, email=juilibaviskar@ashikagroup.com, c=IN Date: 2026.03.25 16:41:42 +05'30'

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Details of Associates

Sl No	Name	CIN/ LLPIN	Registration Number
1	Ashika Credit Capital Ltd	L67120WB1994PLC062159	RBI registered NBFC bearing Registration No. 5.2892
2	Ashika Capital Ltd	U30009WB2000PLC091674	Merchant Banker bearing Registration No INM000010536
3	Ashika Stock Broking (IFSC) Pvt Ltd	U65929GJ2016PTC094597	Stock Broker with NSE IFSC & India INX bearing Registration No - INZ000099630
4	Ashika Business Pvt Ltd	U45100WB2004PTC098055	NA
5	Ashika Properties Pvt Ltd	U70101WB2005PTC102582	NA
6	Ashika Global Securities Pvt Ltd	U65929WB1995PTC069046	RBI registered NBFC bearing Registration No. B.05.00008
7	Ashika Logistics Pvt Ltd	U67200WB2004PTC098054	NA
8	Ashika Global Finance Pvt Ltd	U01132WB1994PTC066087	RBI registered NBFC bearing Registration No. B-05.5583
9	Ashika Entercon Pvt Ltd	U70103WB2017PTC220511	NA
10	Ashika Wealth Management Pvt Ltd	U65999WB2018PTC227019	NA
11	Ashika Investment Managers Pvt Ltd	U65929MH2017PTC297291	Investment Manger to Ashika Alternative Investments, a Category III AIF bearing Registration No. IN/AIF3/20-21/0811
12	Ashika Commodities & Derivatives Pvt Ltd	U51909WB2003PTC096985	NA
13	Puja Sales Promotion Pvt Ltd	U51109WB1993PTC059596	NA
14	Dhara Dealers Pvt Ltd	U52190WB2011PTC169226	NA
15	Yaduka Financial Services Ltd	U51109WB2007PLC117012	RBI registered NBFC bearing Registration No. N.05.06760
16	Ashika Minerals India LLP	AAR-7627	NA
17	Ashika Vyapaar LLP	AAE-3310	NA
18	Ashika Ventures LLP (formerly known as Ashika Rise Realty LLP)	AAO-9947	NA